

Chapter 36

Image Signal Processor (ISP)

36.1 Introduction

ESP32-P4 includes an image signal processor (ISP), which is a pipeline composed of various image processing algorithms.

ISP reads image data from the DVP or MIPI-CSI camera interface, or from the system memory directly, and writes the processed image data to the system memory through VDMA.

The SoC architecture of ISP is shown in Figure 36.1-1. ISP must work with other modules to read and write data and can not work alone.

- When outputting images, ISP uses CSI_Bridge to convert data to AXI interface and stores the images to system memory through the VDMA controller. For more information about VDMA, see Chapter 5 [VDMA Controller \(VDMA\)](#)
- When the input image comes from the system memory, ISP must acquire the image data through VDMA
- When the input image comes from the MIPI-CSI camera interface,
 - ISP must acquire the image data through CSI HOST. For details, see Chapter 40 [MIPI CSI](#)
 - CSI_CM can preprocess it to perform 8-bit or 16-bit bit-swapping, color-channel reordering for RGB888, RGB565, or YUV422, downsample RGB888 to RGB565, or downsample YUV422 to YUV420
- When the input image comes from the DVP camera interface, ISP acquire the image data through the DVP IO. For details, see Chapter 9 [GPIO Matrix and IO MUX](#)

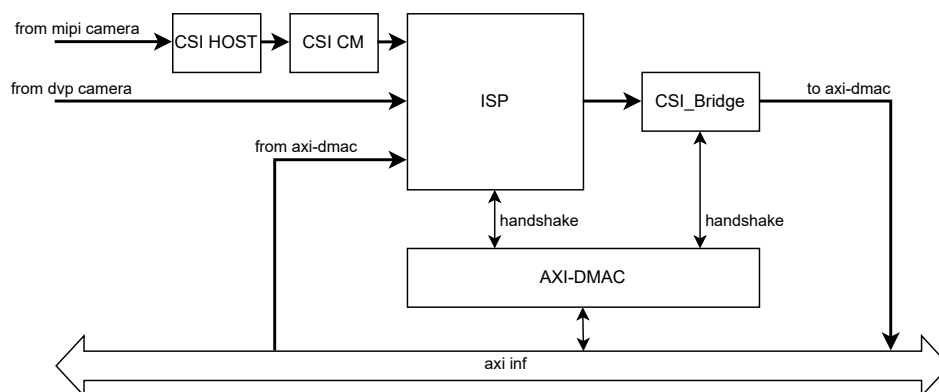


Figure 36.1-1. ESP32-P4 ISP SoC Architecture

36.2 Terminology

MIPI-CSI	Camera serial interface, a high-speed serial interface for cameras compliant with MIPI specifications
Image Interface	Image data interface, the output interface of CSI HOST Classified into Image Interface 32 and Image Interface 64 based on bit width
DVP	Digital video parallel interface, generally composed of vsync, hsync, de, and data signals
RAW	Unprocessed data directly output from an image sensor, typically divided into R, Gr, Gb, and B channels Classified into RAW8, RAW10, RAW12, etc., based on bit width
RGB	Colored image format composed of red, green, and blue colors Classified into RGB888, RGB565, etc., based on the bit width of each color
YUV	Colored image format composed of luminance and chrominance Classified into YUV444, YUV422, YUV420, etc., based on the data arrangement

36.3 Feature List

ISP supports the following features:

- maximum resolution: 1920 x 1080
- three input channels: MIPI-CSI, DVP, and VDMA
- input formats: RAW8, RAW10, and RAW12
- output formats: RAW8, RGB888, RGB565, YUV422, and YUV420
- pipeline features:
 - Black level correction (BLC)
 - Defect pixel correction (DPC)
 - Bayer filter (BF)
 - Lens shading correction (LSC)
 - Demosaic
 - White balance gain (WBG)
 - Color correction matrix (CCM)
 - Gamma correction
 - RGB2YUV
 - Sharpen
 - Contrast/hue/saturation/luminance adjustment (COLOR)
 - YUV_limit
 - YUV2RGB
 - Crop
 - Automatic exposure statistics (AE)
 - Automatic focus statistics (AF)

- Automatic white balance statistics (AWB)
- Histogram statistics (HIST)

36.4 Architectural Overview

ISP architecture is shown in Figure 36.4-1, consisting of ISP_Header, ISP_Pipeline, ISP_Tail, and CSI_Bridge.

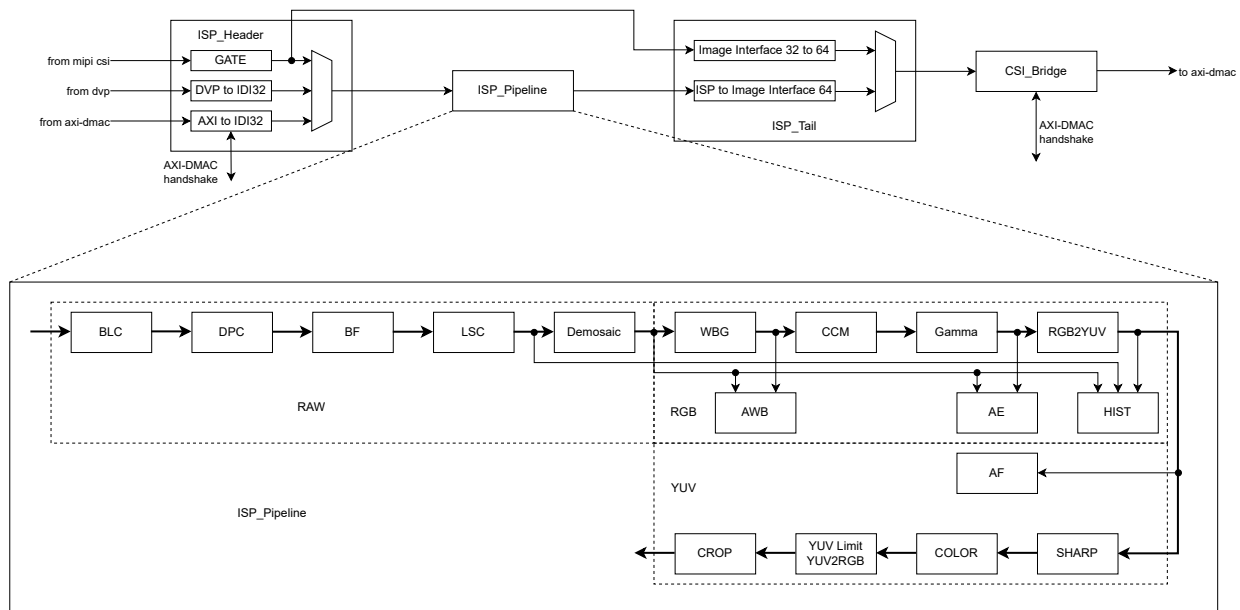


Figure 36.4-1. ISP IP Architecture

- ISP_Header selects data from various input sources based on the configuration and buffers it. Then it transmits the buffered data to ISP_Pipeline in a standardized format.
- ISP_Pipeline serves as the core module of the ISP, containing all the algorithms for image processing.
- ISP_Tail has two input sources, MIPI-CSI and ISP_Pipeline. When ISP is disabled, Image Interface 32 data from MIPI-CSI is converted into Image Interface 64 format by ISP_Tail and transmitted to CSI_Bridge. When ISP is enabled, the output of ISP_Pipeline is cropped and converted into Image Interface 64 format by ISP_Tail, and then transmitted to CSI_Bridge.
- As the output of the ISP, CSI_Bridge transmits the images from ISP to the system memory through VDMA.

36.5 Functional Description

36.5.1 ISP_Header

ISP_Header serves as the input of ISP. The main features are:

- Select image data as input source based on the configuration of [ISP_IN_SRC](#)
- Buffer image data, and convert images to the format required by ISP_Pipeline based on the configured type and size of image data

- When MIPI-CSI is the input, set the input type via [ISP_DATA_TYPE](#)
- When DVP is the input, set [ISP_DATA_TYPE](#) to 0, then set the data type via [ISP_CAM_DATA_TYPE](#)
- When VDMA is the input, set the input type via [ISP_DATA_TYPE](#), and configure [ISP_DMA_DATA_TYPE](#) accordingly

36.5.2 ISP_Pipeline

ISP_Pipeline is the core of ISP, housing all the image processing algorithms. Each algorithm module can be enabled or disabled via the corresponding bit of [ISP_CNTL_REG](#). Except for modules related to color space conversion (demosaic, RGB2YUV, and YUV2RGB), other modules can be enabled or disabled dynamically without the need to shut down the ISP for reconfiguration. Moreover, once enabled, each algorithm module generates an end-of-frame interrupt, capable of triggering an interrupt after processing each image frame.

36.5.2.1 Black Level Correction (BLC)

This module is used to subtract the black level offset of the sensor. The black level offset to be subtracted for each Bayer channel is configured via [ISP_BLC_Rx_VALUE](#). With the black level offset subtracted, the pixel range will be reduced, which can be stretched back to 0-255 via [ISP_BLC_Rx_STRETCH](#).

There are four Bayer channels in total. The mapping between channel and index is as follows:

- R0 refers to the top-left channel
- R1 refers to the top-right channel
- R2 refers to the bottom-left channel
- R3 refers to the bottom-right channel

36.5.2.2 Defective Pixel Correction (DPC)

This module removes defective pixels from the input image. DPC supports both dynamic correction and static correction, which are controlled by [ISP_STA_EN](#) and [ISP_DYN_EN](#), respectively. The two correction modes can be enabled simultaneously.

Dynamic correction automatically detects defective pixels based on the input pixel values and replaces them with the median value. Two defective pixel detection algorithms are supported:

- **Algorithm 0:** Compute the minimum value *min8* and the maximum value *max8* among the eight neighboring pixels of the current pixel. If the current pixel value is greater than *max8* + [ISP_DPC_THRESHOLD_H](#), or less than *min8* - [ISP_DPC_THRESHOLD_L](#), the pixel is classified as defective.
- **Algorithm 1:** Let the current pixel value be *pix*. Compute the mean value *est* of the eight neighboring pixels, and the absolute difference *dif* = |*est* - *pix*|. Define *avg* = (sum of the eight neighbors + 8 × *pix*) / 16. Detection is performed in two steps:
 - **Step 1:** Determine the maximum value *max8* among the eight neighboring pixels. If *pix* < *max8* × [ISP_DPC_THRESHOLD_H](#) and *pix* > *max8* × [ISP_DPC_THRESHOLD_L](#), the pixel is considered non-defective; otherwise, proceed to Step 2.
 - **Step 2:** The pixel is classified as defective if either of the following conditions is met:

- * $dif > avg \times ISP_DPC_FACTOR_DARK$ and $est \geq pix$
- * $dif > (255 - avg) \times ISP_DPC_FACTOR_BRIG$ and $est < pix$

Static correction is implemented using a defective pixel coordinate lookup table (LUT). Pixels whose coordinates are stored in the LUT are replaced with the median value. The defective pixel coordinates must be obtained through static calibration. During calibration, neither static nor dynamic correction can be enabled.

The calibration procedure is as follows:

- Black image defective pixel calibration
 - Set `ISP_DPC_BLACK_EN` = 1
 - Set `ISP_DPC_CHECK_EN` = 1
 - Set `ISP_DPC_EN` = 1
 - Input a black image and wait for the calibration to complete
 - Set `ISP_DPC_CHECK_EN` = 0
 - Software reads all defective pixel coordinates for the black image from the LUT registers
- White image defective pixel calibration
 - Set `ISP_DPC_BLACK_EN` = 0
 - Set `ISP_DPC_CHECK_EN` = 1
 - Set `ISP_DPC_EN` = 1
 - Input a white image and wait for the calibration to complete
 - Set `ISP_DPC_CHECK_EN` = 0
 - Software reads all defective pixel coordinates for the white image from the LUT registers
- Software merges the defective pixel coordinates obtained from the black and white images, removes duplicates, and writes the final list back to the LUT

The LUT read and write procedure is described in Section 36.5.2.4. By setting `ISP_LUT_NUM` = 1, the defective pixel coordinate LUT for DPC is selected. This LUT has a depth of 512, allowing up to 512 defective pixel coordinates to be stored.

36.5.2.3 Bayer Filter (BF)

This module is designed for denoising input images in the Bayer domain. The denoising strength can be adjusted using `ISP_SIGMA`, and the denoising template can be fine-tuned using `ISP_GAU_TEMPLATExx`.

36.5.2.4 Lens Shading Correction (LSC)

This module is utilized for lens shading correction in the Bayer domain. It corrects the R, Gr, Gb, and B channels independently by dividing each channel into a 32 x 32 grid (equivalent to a 64 x 64 division for the entire image) and stores the correction coefficients for each channel in a lookup table (LUT).

Before using this module, calibration is necessary to obtain the LUT correction coefficients. Specifically, first we capture a white image in a scene with uniform ambient lighting. Divide the image into a 32 x 32 grid for

each channel, and use the luminance at the center of the image as a reference. Calculate the luminance gain at each grid vertex based on this reference luminance. For grids extending beyond the image boundaries on the right or bottom, calculate the gain at the image boundary points. Then, use a linear relationship based on the gain of the previous point to calculate the gain at the next grid boundary. Finally, store these gains in the LUT in a left-to-right, top-to-bottom order to complete the initialization.

In the context of a 1920 x 1080 resolution image, each channel operates at 960 x 540 resolution and is divided into a 32 x 32 grid, consisting of 30 grids horizontally and 17 grids vertically. Assuming the image's luminance center aligns with its geometric center, we derive the reference luminance *ref_lum* from the average luminance of pixels around coordinates (480, 270). Moving methodically from left to right and top to bottom, derive the actual pixel luminance *act_lum* for each grid vertex. Using the formula $gain = ref_lum / act_lum$, we compute the correction coefficient, which is then stored sequentially in the LUT.

In cases where grids extend beyond the image boundaries, such as the vertical coordinate of the last grid reaching 543 (beyond the maximum vertical coordinate of 539), we leverage the correction coefficients *gain_0* at coordinate 511 and *gain_1* at coordinate 539. Then the gain for coordinate 543 can be calculated based on $gain = gain_0 + (gain_1 - gain_0) / (539 - 511) * (543 - 511)$.

LSC employs two LUTs for storing gains related to specific color channels: Gb and B are stored in one LUT, while R and Gr are stored in the other. The gain values are represented as 10-bit fixed-point numbers, with the high 2 bits representing the integer part and the low 8 bits the fractional part. The storage order within the LUTs is detailed in Table 36.5-1. The LUTs can be accessed through specific registers:

- A write operation writes the data from [ISP_LUT_WDATA_REG](#) into the LUT.
- A read operation stores the data read from the LUT into [ISP_LUT_RDATA_REG](#).
- Trigger LUT operations via [ISP_LUT_CMD_REG](#). Each write to this register initiates a read or write operation on the LUT. It is required to write to all the fields within this register at the same time.

Table 36.5-1. Gain Storage Sequence in LSC LUTs

LUT sel	bit [19:10]	bit [9:0]
LUT_Gb_B	Gb	B
LUT_R_Gr	R	Gr

Before enabling LSC, [ISP_LSC_XTABLESIZE](#) should be configured to set the number of horizontal correction coefficients. The number is determined by $ISP_LSC_XTABLESIZE = fix((line_width - 1) / 2 / 32) + 2$, where *fix()* denotes rounding down. For example, if the input RAW image has a horizontal width of 1920 pixels, then $ISP_LSC_XTABLESIZE = fix((1920 - 1) / 2 / 32) + 2 = 29 + 2 = 31$.

36.5.2.5 Demosaic

The demosaicing process converts Bayer images to RGB888 images. The conversion effect can be fine-tuned using [ISP_DEMOSAIC_GRAD_RATIO](#). This register field comprises 2 integer bits and 4 fractional bits, with a default value of 1.0.

36.5.2.6 White Balance Gain (WBG)

This module performs white balance adjustment on the input image. The adjustment is achieved by multiplying the pixel value of each color channel by its corresponding [ISP_WBG_R/B/G](#) gain coefficient. Each

gain coefficient consists of a 4-bit integer part and an 8-bit fractional part.

36.5.2.7 Color Correction Matrix (CCM)

CCM is a correction matrix for image pixels that can adjust RGB888 pixels, including white balance adjustments. The relationship between the output R', G', B' and the input R, G, B is described by the following matrix. The parameters RR, RG, RB, GR, GG, GB, BR, BG, and BB can be configured with [ISP_CCM_xx](#). Each parameter is a 13-bit fixed-point number, where bit [12] is the sign bit, bits [11:0] represent the absolute value of the number, bits [11:8] the integer part, and bits [7:0] the fractional part.

$$\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} RR & RG & RB \\ GR & GG & GB \\ BR & BG & BB \end{bmatrix} \times \begin{bmatrix} R \\ G \\ B \end{bmatrix} \quad (36.1)$$

36.5.2.8 Gamma Correction

This module is used to adjust the gamma curve of an image. The gamma curve should be configured before gamma correction is enabled. Gamma curves can be configured using registers, with each R, G, B channel having an independent curve consisting of 16 sampling points. For the X component (input value), the values are indirectly obtained through [ISP_GAMMA_R/G/B_Xn](#) (where n ranges from 00 to 0F). The Y component (output value) can be directly set using [ISP_GAMMA_R/G/B_Yn](#) (where n ranges from 00 to 0F). Once configuration is complete, writing 1 to [ISP_GAMMA_UPDATE](#) applies the gamma curve settings. The gamma curve is illustrated in Figure 36.5-1.

For the X component, taking the R channel as an example,

$ISP_GAMMA_R_X(n) = \log_2(X(n) - X(n-1))$. This means the nth X coordinate value is obtained by adding 2 to the power of $ISP_GAMMA_R_X(n)$ to the (n-1)th X coordinate value. Specifically, for the last coordinate, $X(0F) = 2^{ISP_GAMMA_R_X(0F)} + X(0E) - 1$. Since RGB888 in ISP uses 8-bit integers, only the configuration where $X(0F) = 255$ is considered a valid configuration.

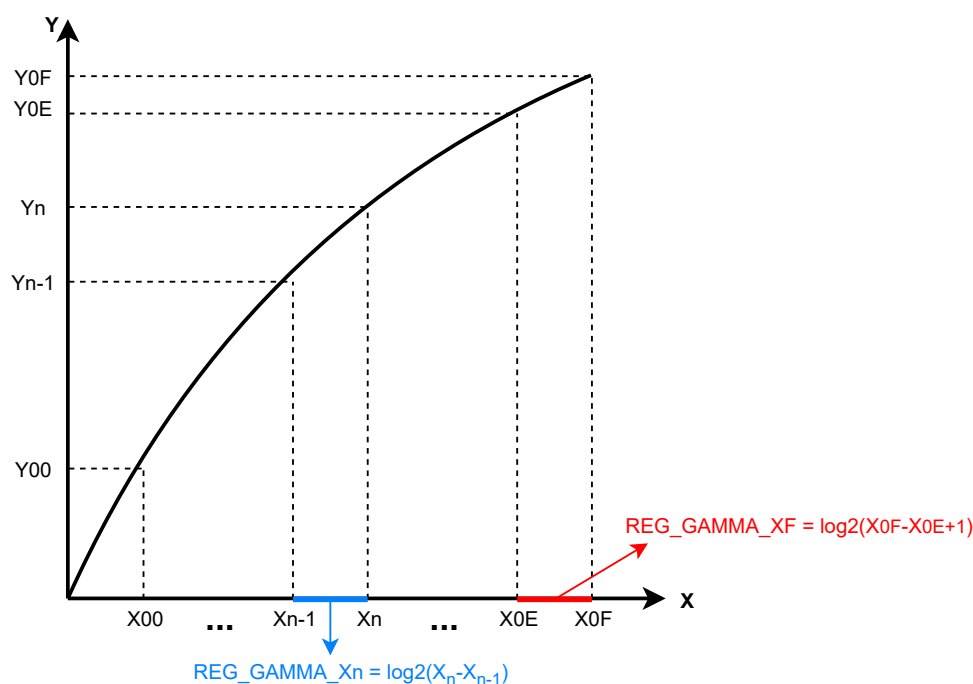


Figure 36.5-1. Gamma Curve

36.5.2.9 RGB2YUV

The RGB2YUV module converts RGB888 pixels into full-range YUV422 format. The conversion protocol can be configured with [ISP_YUV_MODE](#).

36.5.2.10 Sharpen

The sharpen module is used to sharpen images. This module separates the low-frequency and high-frequency components of the image using a low-pass filter template (such as a mean template or Gaussian template) via [ISP_SHARP_FILTER_COEFF](#). For low-frequency components, no processing will be done. For high-frequency components, each pixel undergoes the following processing to achieve sharpening:

- Pixels with frequencies below [ISP_SHARP_THRESHOLD_LOW](#) are set to 0.
- Pixels with frequencies above [ISP_SHARP_THRESHOLD_LOW](#) but below [ISP_SHARP_THRESHOLD_HIGH](#) are multiplied by the coefficient [ISP_SHARP_AMOUNT_LOW](#).
- Pixels with frequencies above [ISP_SHARP_THRESHOLD_HIGH](#) are multiplied by the coefficient [ISP_SHARP_AMOUNT_HIGH](#).

At the end of each frame, the maximum value of high-frequency component pixels in that frame can be obtained by reading [ISP_SHARP_GRADIENT_MAX](#). This value can serve as a reference for configuring settings for the next frame.

36.5.2.11 Contrast/Hue/Saturation/Luminance Adjustment (COLOR)

This module is designed to modify the contrast, saturation, hue, and luminance of an image. Contrast, saturation, and luminance are configured through [ISP_COLOR_CONTRAST](#), [ISP_COLOR_SATURATION](#), and [ISP_COLOR_BRIGHTNESS](#), respectively. Hue adjustment is achieved by combining [ISP_COLOR_HUE_H](#) and [ISP_COLOR_HUE](#) to form a 9-bit control value.

36.5.2.12 YUV_Limit and YUV2RGB

The YUV_Limit module converts full-range YUV422 pixels to limited-range YUV422 pixels. This module is activated only when RGB2YUV is enabled, YUV2RGB is disabled, and [ISP_YUV_RANGE](#) is set to 1. With this configuration, the ISP output type is either YUV422 or YUV420.

YUV2RGB converts full-range YUV422 pixels to RGB888 format. The conversion protocol can be configured with [ISP_YUV_MODE](#).

Note:

For full-range YUV, Y, U, and V channels range from 0 to 255.

For limited-range YUV, Y channel ranges from 16 to 235, and U and V channels range from 16 to 240.

36.5.2.13 Cropping (CROP)

The CROP module performs image cropping. The start and end coordinates in the X and Y directions are configured using [ISP_CROP_X_START](#), [ISP_CROP_X_END](#), [ISP_CROP_Y_START](#), and [ISP_CROP_Y_END](#), respectively. The start coordinates must be even, and the end coordinates must be odd.

36.5.2.14 Automatic Exposure Statistics (AE)

Auto Exposure (AE) uses 25 statistical sub-windows for image luminance statistics, forming a 5 x 5 grid of statistical window. The size of these windows is set using the register [ISP_AE_BX/BY_REG](#). Each sub-window is numbered as shown in Figure 36.5-2. The average luminance of each sub-window, represented as an 8-bit value, can be read with [ISP_AE_Bxx_MEAN](#). These statistics can be utilized to implement automatic exposure algorithms for cameras.

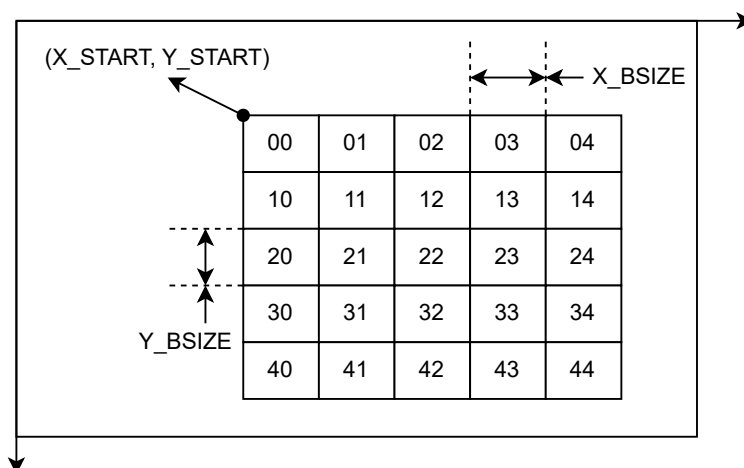


Figure 36.5-2. Sub-Window Numbering

AE utilizes two data sampling points: the output of the [Demosaic](#) module and the output of the [Gamma](#) module. These sampling points can be configured using [ISP_AE_SELECT](#).

AE also includes a simple luminance monitoring module, implemented either by hardware or software. This module calculates the average luminance of all sub-windows every [ISP_AE_MONITO_RPERIOD](#) frames. If the average luminance falls outside the thresholds set by [ISP_AE_MONITOR_TH/TL](#), interrupt for AE luminance

monitoring will be triggered. Note that the luminance monitoring module does not support setting different weights for individual sub-windows. Therefore, if you wish to implement monitoring with different weights (for instance, prioritize highlights), you need to perform software-triggered manual statistics and process the statistical data accordingly.

36.5.2.15 Automatic Focus Statistics (AF)

AF gathers luminance and sharpness information within specified statistical windows based on the output of the [sharpen](#) module. There are 3 independent statistical windows, and the size and position of each window can be configured using [ISP_AF_H/VSCALE_A/B/C_REG](#). The module supports automatic and manual trigger modes for statistics collection. The results of luminance and sharpness statistics for each window can be accessed using [ISP_AF_LUMA/B/C](#) and [ISP_AF_SUMA/B/C](#). These statistics can be used to implement automatic focus algorithms for cameras.

AF also includes a focus scene monitoring module, which users can choose to implement hardware-based monitoring with this module or implement via software. This module evaluates cumulative changes in luminance and sharpness across all statistical windows as criteria. It performs statistics every [ISP_AF_ENV_PERIOD](#) frames. If consecutive statistics show a difference from baseline values for luminance and sharpness that exceeds the specified threshold, an interrupt is triggered indicating a change in the focus scene that requires refocusing. The baseline values are automatically updated after each automatic or manually triggered statistics collection. There are three methods for setting the threshold:

- Directly set the luminance and sharpness thresholds using [ISP_AF_ENV_USER_THRESHOLD_LUM](#) and [ISP_AF_ENV_USER_THRESHOLD_SUM](#)
- If [ISP_AF_ENV_USER_THRESHOLD_SUM](#) is 0, use the corresponding baseline sharpness value multiplied by [ISP_AF_ENV_THRESHOLD](#) (4 fractional bits) as the sharpness threshold
- If [ISP_AF_ENV_USER_THRESHOLD_LUM](#) is 0, use the corresponding baseline luminance value multiplied by [ISP_AF_ENV_THRESHOLD](#) (4 fractional bits) as the luminance threshold

36.5.2.16 Automatic White Balance Statistics (AWB)

AWB gathers white balance information for specified windows containing white patches. The coordinates of the window can be set using [ISP_AWB_H/VSCALE_REG](#). White patches are filtered based on luminance limits [ISP_AWB_MAX/MIN_LUM](#), *R/G* limits [ISP_AWB_MAX/MIN_RG](#), and *B/G* limits [ISP_AWB_MAX/MIN_BG](#).

The statistics include the number of white patches and the accumulated values of the R, G, B components for all white patches. These statistics can be obtained using [ISP_AWBO_WHITE_CNT](#) and [ISP_AWBO_ACC_R/G/B](#). They can be used to implement automatic white balance algorithms in cameras.

In addition, AWB supports dividing the window into 5×5 sub-windows, which are configured via [ISP_AWB_X_START](#), [ISP_AWB_X_BSIZE](#), [ISP_AWB_Y_START](#), and [ISP_AWB_Y_BSIZE](#). The sub-window range must not exceed the range of the main window. The statistics results are written to the AWB LUT and can be accessed by setting [ISP_LUT_NUM](#) = 2. The detailed access procedure can be found in Section 36.5.2.4. The statistics are stored in the order of sub-windows from left to right and from top to bottom. Each sub-window occupies four consecutive addresses, which store, from low to high addresses, the number of white patches, and the accumulated values of the R, G, B components for white patches.

AWB relies on two data sampling points: the output of [Demosaic](#) and the output of [CCM](#), which can be configured with [ISP_AWB_SAMPLE](#).

36.5.2.17 Histogram Statistics (HIST)

HIST calculates brightness information for the image through statistical analysis. This module can define a statistical window, which can be divided into 5 x 5 sub-windows similar to AE. The configuration is done using registers [ISP_HIST_OFFSETS_REG](#) and [ISP_HIST_SIZE_REG](#). The weight of each sub-window can be independently adjusted with [ISP_HIST_WEIGHT_xx](#).

The statistics are presented as a histogram with 16 intervals. The x-axis of the histogram is defined using [ISP_HIST_SEG_x_x](#), and the final statistics can be retrieved using [ISP_HIST_BIN_x](#).

HIST involves three data sampling points: [BF](#), [demosaic](#), and [RGB2YUV](#), corresponding to RAW, RGB, and YUV data sampling points respectively. These can be configured using [ISP_HIST_MODE](#). When the sampling point is set to RGB, RGB weighting coefficients can be adjusted via [ISP_HIST_COEFF_R/G/B](#).

36.5.3 ISP_Tail

ISP_Tail is the post-processing section that handles data from two input sources, ultimately outputting in Image Interface 64 format to CSI_Bridge.

When the ISP is turned off, ISP_Tail receives Image Interface 32 data from the MIPI CSI HOST, converts it to Image Interface 64 format, and outputs it to CSI_Bridge. The byte order of the input Image Interface 32 data can be adjusted using [ISP_BYTE_ENDIAN_ORDER](#).

When the ISP is turned on, ISP_Tail receives data from ISP_Pipeline and processes them before outputting to CSI_Bridge. This processing includes:

- Converting RGB888 to RGB565 based on register configurations
- Converting YUV422 to YUV420 based on register configurations
- Converting data from ISP_Pipeline to Image Interface 64 format

36.5.4 Sequence Control

For modules involving matrix operations, it is necessary to buffer multiple lines of data before starting operations. In such cases, it is crucial for the module to generate an output sequence when the last few lines of image data are buffered. This output sequence can be controlled using

[ISP_BF/DEMOSAIC/SHARP_TAIL_PIXEN_PULSE_TH/TL](#), to prevent buffer overflow in the CSI_Bridge due to excessive speed. Please exercise caution when setting this register too large, as it may disrupt the normal data sequence. TH must be smaller than [ISP_HADR_NUM](#)-1, and it is recommended to set [ISP_BF/DEMOSAIC/SHARP_TAIL_PIXEN_PULSE_TH/TL](#) to a value less than or equal to 8.

36.5.5 Shadow Registers

Some ISP modules support shadow registers. The supported modules include:

- BLC
- DPC

- BF
- WBG
- CCM
- GAMMA
- SHARP
- COLOR

For the GAMMA module, the shadow register update method is fixed and must strictly follow the procedure described in Section 36.5.2.8. For all other modules, the update method can be configured via [ISP_SHADOW_UPDATE_SEL](#).

36.5.6 CSI_CM

For MIPI CSI input, the CSI_CM module can perform simple data conversions by configuring [CSI_BRIG_HOST_CM_CTRL_REG](#), including:

- Swapping high and low 8-bit data
- Swapping high and low 16-bit data
- Adjusting RGB888 color channel order (pixels per line must be a multiple of 4)
- Down-sampling from RGB888 to RGB565 (pixels per line must be a multiple of 4)
- Adjusting RGB565 color channel order (pixels per line must be a multiple of 2)
- Adjusting YUV422 color channel order (pixels per line must be a multiple of 2)
- Down-sampling from YUV422 to YUV420 (pixels per line must be a multiple of 16)

The module structure is shown in Figure 36.5-3.

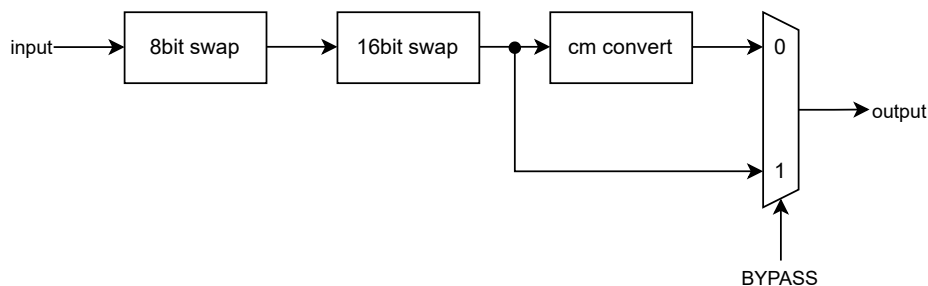


Figure 36.5-3. ESP32-P4 CSI CM Architecture

36.5.7 CSI_Bridge

CSI_Bridge serves as the interface between the ISP and the system memory. This module utilizes VDMA to transmit data from the ISP into the system memory. The image can be transferred either as a single block or divided into multiple blocks, which can be configured via [CSI_BRIG_DMA_BURST_LEN](#) and [CSI_BRIG_DMABLK_SIZE](#).

- [CSI_BRIG_DMA_BURST_LEN](#) configures the amount of 64-bit data in a VDMA burst transfer.

- [CSI_BRIG_DMABLK_SIZE](#) configures the number of VDMA bursts in a VDMA block transfer. If an image frame is transferred with only one block, and this block contains n VDMA bursts, then configure this field to a value greater than n . It is recommended to set it to the maximum value, 0x1FFF.

36.5.8 Color Mode and Byte Order

36.5.8.1 Output Pixel Layout Format

When the ISP is disabled, ISP_Tail directly receives Image Interface 32 data from the MIPI CSI HOST. The pixel layout can be referenced from Chapter 40 [MIPI CSI](#). In this case, the Image Interface 32 data output from CSI HOST can go through two levels of byte ordering in ISP_Tail and CSI_Bridge, refer to [Byte Order](#) for details.

After the ISP is enabled and the data has been processed by the ISP_Pipeline, the layout format of the output pixel is as shown in Table 36.5-2.

Table 36.5-2. ISP Output Pixel Layout Format

Color Mode	Addr + 7	Addr + 6	Addr + 5	Addr + 4	Addr + 3	Addr + 2	Addr + 1	Addr + 0
RAW8	RAW7[7:0]	RAW6[7:0]	RAW5[7:0]	RAW4[7:0]	RAW3[7:0]	RAW2[7:0]	RAW1[7:0]	RAW0[7:0]
RGB888	G2[7:0]	B2[7:0]	R1[7:0]	G1[7:0]	B1[7:0]	R0[7:0]	G0[7:0]	B0[7:0]
	B5[7:0]	R4[7:0]	G4[7:0]	B4[7:0]	R3[7:0]	G3[7:0]	B3[7:0]	R2[7:0]
	R7[7:0]	G7[7:0]	B7[7:0]	R6[7:0]	G6[7:0]	B6[7:0]	R5[7:0]	G5[7:0]
RGB565	R3[4:0]G3[5:3]	G3[2:0]B3[4:0]	R2[4:0]G2[5:3]	G2[2:0]B2[4:0]	R1[4:0]G1[5:3]	G1[2:0]B1[4:0]	R0[4:0]G0[5:3]	G0[2:0]B0[4:0]
YUV422	Y3[7:0]	V2[7:0]	Y2[7:0]	U2[7:0]	Y1[7:0]	V0[7:0]	Y0[7:0]	U0[7:0]
YUV420 (Odd-numbered rows)	Y4[7:0]	U4[7:0]	Y3[7:0]	Y2[7:0]	U2[7:0]	Y1[7:0]	Y0[7:0]	U0[7:0]
	U10[7:0]	Y9[7:0]	Y8[7:0]	U8[7:0]	Y7[7:0]	Y6[7:0]	U6[7:0]	Y5[7:0]
	Y15[7:0]	Y14[7:0]	U14[7:0]	Y13[7:0]	Y12[7:0]	U12[7:0]	Y11[7:0]	Y10[7:0]
YUV420 (Even-numbered rows)	Y4[7:0]	V4[7:0]	Y3[7:0]	Y2[7:0]	V2[7:0]	Y1[7:0]	Y0[7:0]	V0[7:0]
	V10[7:0]	Y9[7:0]	Y8[7:0]	V8[7:0]	Y7[7:0]	Y6[7:0]	V6[7:0]	Y5[7:0]
	Y15[7:0]	Y14[7:0]	V14[7:0]	Y13[7:0]	Y12[7:0]	V12[7:0]	Y11[7:0]	Y10[7:0]

36.5.8.2 Byte Order

CSI_Bridge can reorder the bytes of the input 64-bit data via [CSI_BRIG_BYTE_ENDIAN_ORDER](#). ISP_Tail can also reorder the bytes via [ISP_BYTE_ENDIAN_ORDER](#). The byte ordering functionality within ISP_Tail differs from that of CSI_Bridge, as illustrated in Figure 36.5-4.

- ISP_Tail byte ordering: This operation applies to its Image Interface 32 input. It reverses the high and low bytes of the 32-bit input data and then concatenates them to output 64-bit data. This step only functions when the ISP is disabled and is mainly used for processing Image Interface 32 YUV422 and YUV420 input types.
- CSI_Bridge byte ordering: This operation acts on the 64-bit input data (i.e., the output data from ISP_Tail) by reversing its high and low bytes.

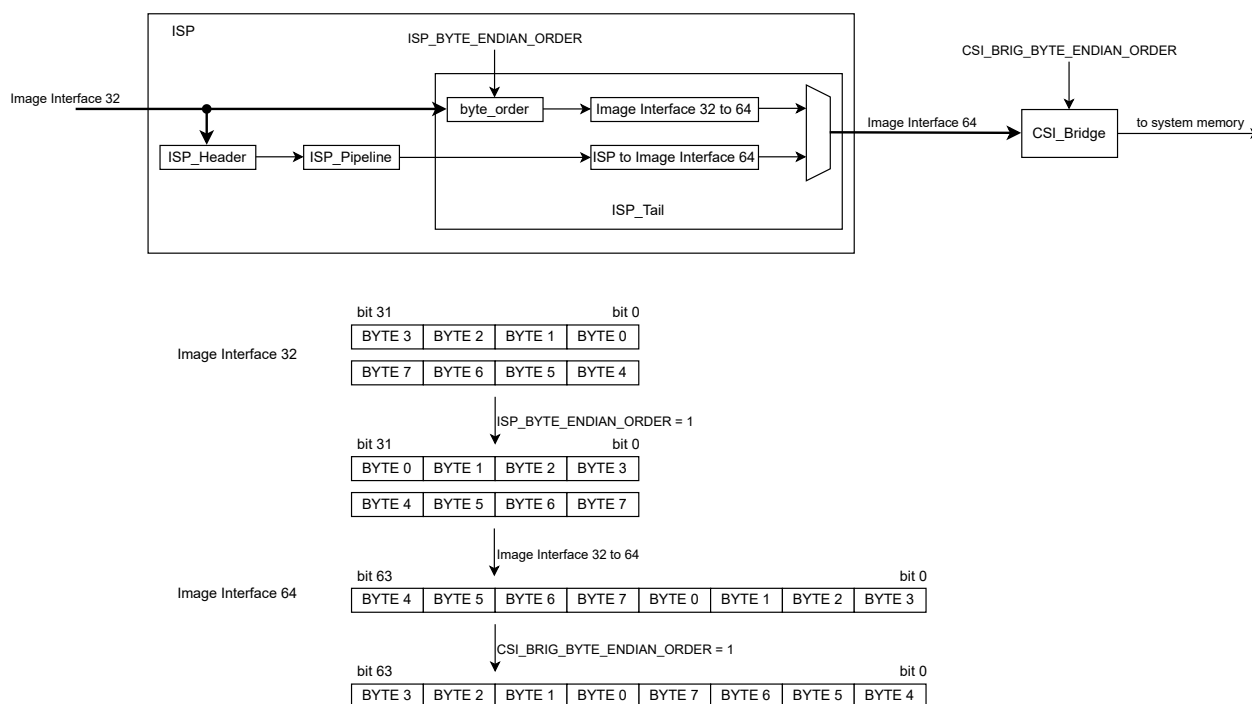


Figure 36.5-4. Byte Ordering

36.6 Interrupts

ESP32-P4's ISP can generate the following interrupt signals that will be sent to the [Interrupt Matrix](#).

- `isp_interrupt`
- `csi_bridge_interrupt`

There are several internal interrupt sources from ISP that can generate the above interrupt signal. The interrupt sources from ISP are listed with their trigger conditions and the resulted interrupt signal in [Table 36.6-1](#).

Table 36.6-1. ISP's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
ISP_CROP_ERR_INT	When the crop module window is configured incorrectly	<code>isp_interrupt</code>
ISP_WBG_FRAME_INT	After module WBG processes a frame	<code>isp_interrupt</code>
ISP_CROP_FRAME_INT	After module crop processes a frame	<code>isp_interrupt</code>
ISP_HEADER_IDI_FRAME_INT	After MIPI-CSI inputs a frame. This interrupt is not affected by whether ISP_MIPI_DATA_EN is enabled	<code>isp_interrupt</code>
ISP_TAIL_IDI_FRAME_INT	After ISP_Tail outputs a frame	<code>isp_interrupt</code>
ISP_YUV2RGB_FRAME_INT	After module YUV2RGB processes a frame	<code>isp_interrupt</code>
ISP_COLOR_FRAME_INT	After module contrast/hue/saturation/luminance adjustment processes a frame	<code>isp_interrupt</code>
ISP_SHARP_FRAME_INT	After module sharpen processes a frame	<code>isp_interrupt</code>

ISP_RGB2YUV_FRAME_INT	After module RGB2YUV processes a frame	isp_interrupt
ISP_GAMMA_FRAME_INT	After module gamma correction processes a frame	isp_interrupt
ISP_CCM_FRAME_INT	After module CCM processes a frame	isp_interrupt
ISP_DEMOSAIC_FRAME_INT	After module demosaic processes a frame	isp_interrupt
ISP_BF_FRAME_INT	After module BF processes a frame	isp_interrupt
ISP_DPC_FRAME_INT	After module DPC processes a frame	isp_interrupt
ISP_LSC_FRAME_INT	After module LSC processes a frame	isp_interrupt
ISP_BLC_FRAME_INT	After module BLC processes a frame	isp_interrupt
ISP_FRAME_INT	After ISP_Pipeline outputs a frame	isp_interrupt
ISP_HIST_FDONE_INT	After module HIST processes a frame	isp_interrupt
ISP_AWB_FDONE_INT	After module AWB processes a frame	isp_interrupt
ISP_AF_ENV_INT	When module AF detects a change in the focus scene	isp_interrupt
ISP_AF_FDONE_INT	After module AF processes a frame	isp_interrupt
ISP_AE_FRAME_DONE_INT	After module AE processes a frame	isp_interrupt
ISP_AE_MONITOR_INT	When module AE detects a change in the luminance	isp_interrupt
ISP_GAMMA_XCOORD_ERR_INT	When the GAMMA parameter is set incorrectly, the X length in ISP_GAMMA_R/G/B_X00-OF is not 256	isp_interrupt
ISP_DPC_CHECK_DONE_INT	After DPC static calibration processes a frame	isp_interrupt
ISP_MIPI_HNUM_UNMATCH_INT	When the width HNUM settings do not match the actual input from MIPI-CSI	isp_interrupt
ISP_DATA_TYPE_SETTING_ERR_INT	When DATA_TYPE settings do not match the actual input data	isp_interrupt
ISP_HVNUM_SETTING_ERR_INT	When the width HNUM and the height VNUM settings are incorrect: Image dimensions must be even; with the RAW10 input, HNUM must be a multiple of 4	isp_interrupt
ISP_BUF_FULL_INT	When ISP_Header buffer is full	isp_interrupt
ISP_ASYNC_FIFO_OVF_INT	When asynchronous FIFO of ISP_Header is overflowed	isp_interrupt
ISP_DATA_TYPE_ERR_INT	When illegal DATA_TYPE is used for input data (not RAW8/RAW10/RAW12)	isp_interrupt
CSI_BRIG_DMA_CFG_HAS_UPDATED_INT	When VDMA configuration of CSI_Bridge is updated	csi_bridge_interrupt
CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT	When asynchronous FIFO of CSI_Bridge is overflowed	csi_bridge_interrupt
CSI_BRIG_CSI_BUF_OVERRUN_INT	When CSI_Bridge buffer is overflowed	csi_bridge_interrupt

CSI_BRIG_DISCARD_INT	When any frame is discarded in CSI_Bridge	csi_bridge_interrupt
CSI_BRIG_VADR_NUM_LT_INT	When CSI_BRIG_VADR_NUM is less than the actual value	csi_bridge_interrupt
CSI_BRIG_VADR_NUM_GT_INT	When CSI_BRIG_VADR_NUM is greater than the actual value	csi_bridge_interrupt

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [36.8 Register Summary](#).

36.7 Programming Procedures

36.7.1 ISP Clock Reset Configuration

1. Enable the bus clock for the ISP and CSI_Bridge modules;
 - Set [HP_SYS_CLKRST_CSI_BRG_SYS_CLK_EN](#) to 1 to activate the clock for ISP and CSI_Bridge
2. Configure the working clock for ISP;
 - Choose the clock source via [HP_SYS_CLKRST_ISP_CLK_SRC_SEL](#)
 - Configure the clock divisor via [HP_SYS_CLKRST_ISP_CLK_DIV_NUM](#)
 - Set [HP_SYS_CLKRST_ISP_CLK_EN](#) to 1 to activate the ISP working clock
3. Reset ISP and CSI_Bridge modules.
 - Set [HP_SYS_CLKRST_RST_EN_CSI_BRG](#) to 1 and then 0 to reset ISP and CSI_Bridge

36.7.2 CSI_Bridge Configuration

1. Set the width and height of the image via [CSI_BRIG_HADR_NUM](#) and [CSI_BRIG_VADR_NUM](#). The width is measured in "64-bit" units, and the height is measured in rows;
2. Set the valid range for DATA_TYPE via [CSI_BRIG_DATA_TYPE_MIN](#) and [CSI_BRIG_DATA_TYPE_MAX](#);
3. Specify the burst length via [CSI_BRIG_DMA_BURST_LEN](#). Ensure that this value matches the SRC_MSIZ (length per burst) and SRC_TR_WIDTH (data width) of the corresponding VDMA channel. The data width for CSI_Bridge is measured in "64-bit" units;
4. If the image is transferred in multiple blocks, specify the maximum number of VDMA bursts per block via [CSI_BRIG_DMABLK_SIZE](#);
5. Enable CSI_Bridge by writing 1 to [CSI_BRIG_CSI_BRIG_EN](#).

36.7.3 VDMA Configuration

1. Configure the register corresponding to the channel;
 - Set SRC/DST_MULTBLK_TYPE to 3 (configure with linked list)
 - Configure TT_FC
 - For CSI_Bridge, set TT_FC to 4 (peripheral to memory with peripheral flow control)
 - For memory to ISP, set TT_FC to 6 (memory to peripheral with peripheral flow control)
 - Configure HS_SEL
 - For CSI_Bridge, set HS_SEL_SRC to 0 (peripheral uses hardware handshake), and set HS_SEL_DST to 1 (memory uses software handshake)
 - For memory to ISP, set HS_SEL_SRC to 1 (memory uses software handshake), and set HS_SEL_DST to 0 (peripheral uses hardware handshake)
 - Assign hardware handshake channels
 - For CSI_Bridge, set SRC_PER to 1
 - For memory to ISP, set DST_PER to 2
 - Set SRC/DST_TR_WIDTH to 3 (data width of 64 bits)
2. Establish the linked list based on the configured registers.

Note:

The VDMA configuration outlined in this section covers essential steps. For detailed procedures, refer to Chapter 5 [VDMA Controller \(VDMA\)](#).

Ensure that image dimensions and start addresses are aligned to 8 bytes.

36.7.4 ISP General Configuration

1. Configure the input image size via [ISP_HADR_NUM](#) and [ISP_VADR_NUM](#)
2. Set image arrangement mode in the Bayer domain via [ISP_BAYER_MODE](#);
3. Specify the ISP output data format via [ISP_OUT_TYPE](#);
4. Configure each module in ISP_Pipeline. Refer to the corresponding section in [36.5.2](#) for details;
5. Enable the necessary Pipeline modules by setting the fields in register [ISP_CNTL_REG](#). Ensure that the format conversion-related modules ([Demosaic](#), [RGB2YUV](#), [YUV2RGB](#)) are enabled before enabling the ISP, and that they match the final output format.

36.7.5 Enabling ISP for Image Capture from MIPI-CSI

1. Configure ISP and CSI_Bridge clocks and reset as described in Section [36.7.1](#);
2. Configure CSI_Bridge as described in Section [36.7.2](#);
3. Configure VDMA as described in Section [36.7.3](#);

4. Configure MIPI CSI HOST according to Chapter [40 MIPI CSI](#);
5. Configure ISP general settings as described in Section [36.7.4](#);
6. Select the input source as MIPI-CSI by setting [ISP_IN_SRC](#) to 0;
7. Configure the input data type via [ISP_DATA_TYPE](#);
8. Enable the ISP by setting [ISP_EN](#) to 1;
9. Enable MIPI-CSI data input by setting [ISP_MIPI_DATA_EN](#) to 1.

36.7.6 Enabling ISP for Image Capture from DVP

1. Configure ISP and CSI_Bridge clocks and reset as described in Section [36.7.1](#);
2. Configure CSI_Bridge as described in Section [36.7.2](#);
3. Configure VDMA as described in Section [36.7.3](#);
4. Configure ISP general settings as described in Section [36.7.4](#);
5. Select the input source as DVP by setting [ISP_IN_SRC](#) to 1;
6. Configure the input data type via [ISP_CAM_DATA_TYPE](#), and write 0 to [ISP_DATA_TYPE](#);
7. Enable the ISP by setting [ISP_EN](#) to 1;
8. Configure the data mode for DVP interface via [ISP_CAM_CONF_REG](#);
9. Reset the DVP interface by setting [ISP_CAM_RESET](#) to 1, waiting for a period, then setting it to 0;
10. Write 1 to [ISP_CAM_UPDATE_REG](#);
11. Enable the DVP interface by setting [ISP_CAM_EN](#) to 1;
12. Wait for [ISP_CAM_UPDATE_REG](#) to automatically clear for initialization.

36.7.7 Enabling ISP for Image Capture from VDMA

1. Configure ISP and CSI_Bridge clocks and reset as described in Section [36.7.1](#);
2. Configure CSI_Bridge as described in Section [36.7.2](#);
3. Configure VDMA as described in Section [36.7.3](#);
4. Configure ISP general settings as described in Section [36.7.4](#);
5. Select the input source as VDMA by setting [ISP_IN_SRC](#) to 2;
6. Configure the input data type via [ISP_DMA_DATA_TYPE](#) and [ISP_DATA_TYPE](#);
7. Configure the total amount of raw image data for one frame in "64-bit" units using [ISP_DMA_RAW_NUM_TOTAL](#), which value should equal image width x image height x bits per pixel/64. Then, write 1 to [ISP_DMA_RAW_NUM_TOTAL_SET](#) to activate the configuration;
8. Configure the burst length via [ISP_DMA_BURST_LEN](#), which needs to align with the VDMA configuration;
9. Write 1 to [ISP_DMA_UPDATE_REG](#) to update the VDMA configuration, and wait for the update to complete;
10. Send a frame of data by setting [ISP_DMA_EN](#) to 1.

36.7.8 LSC LUT Configuration

LSC can be configured after the ISP is enabled. Before enabling LSC, initialize the LSC LUT as follows:

1. Write to LUT;
 - Write the data to be stored in the LUT by writing to [ISP_LUT_WDATA_REG](#)
 - Store the data at the specified address by writing to [ISP_LUT_CMD_REG](#)
2. Read from LUT.
 - Retrieve the data stored at the specified address by writing to [ISP_LUT_CMD_REG](#)
 - Obtain the data read from the LUT by reading from [ISP_LUT_RDATA_REG](#)

36.7.9 AE Configuration

AE can be configured after the ISP is enabled as follows:

1. Select sampling points via [ISP_AE_SELECT](#);
2. Set up statistical windows;
 - Set the horizontal starting point with [ISP_AE_X_START](#)
 - Configure the horizontal dimension of each sub-window with [ISP_AE_X_BSIZE](#)
 - Set the vertical starting point with [ISP_AE_Y_START](#)
 - Configure the vertical dimension of each sub-window with [ISP_AE_Y_BSIZE](#)
 - Specify the number of pixels in each sub-window with [ISP_AE_SUBWIN_PIXNUM](#)
 - Set the reciprocal of the number of pixels in each sub-window with [ISP_AE_SUBWIN_RECIP](#) (20 fractional bits)
3. Enable luminance monitoring;
 - If monitoring is required, set [ISP_AE_MONITOR_TH](#) and [ISP_AE_MONITOR_TL](#) to non-zero values
 - Specify the monitoring statistical period in frames via [ISP_AE_MONITOR_PERIOD](#)
4. Enable AE by writing 1 to [ISP_AE_EN](#);
5. Obtain statistics;
 - Clear the completion status by writing 1 to [ISP_AE_FRAME_DONE_INT_CLR](#)
 - Initiate statistics by writing 1 to [ISP_AE_UPDATE](#). Each write to this register resets the frame count for monitoring. It is recommended to disable monitoring before manual statistics and re-enable it after completion to prevent automatic results from overwriting manual results prematurely
 - Wait for the statistics to complete
 - Wait for the ISP interrupt triggered by [AE_FDONE_INT](#)
 - Or query the status using [ISP_AE_FRAME_DONE_INT_RAW](#)
 - Obtain statistics for each sub-window by reading [ISP_AE_Bxx_MEAN](#)
6. Perform luminance monitoring.

- Wait for the ISP interrupt triggered by `AE_MONITOR_INT`
- Or query the status using `ISP_AE_MONITOR_INT_RAW`
- After detecting a change in luminance, disable luminance monitoring, re-initiate camera exposure, and then re-enable monitoring

36.710 AF Configuration

AF can be configured after the ISP is enabled as follows:

1. Set up three statistical windows A, B, and C, ensuring each window does not exceed 1024 x 1024;
 - Set the left boundary using `ISP_AF_LPOINT_A/B/C`
 - Set the right boundary using `ISP_AF_RPOINT_A/B/C`
 - Set the top boundary using `ISP_AF_TPOINT_A/B/C`
 - Set the bottom boundary using `ISP_AF_BPOINT_A/B/C`
2. Set the sharpness threshold via `ISP_AF_THRESHOLD`;
3. Configure focus scene monitoring;
 - If focus scene monitoring is required, set the monitoring period `ISP_AF_ENV_PERIOD` to a non-zero value
 - There are two ways to set the threshold
 - If `ISP_AF_ENV_USER_THRESHOLD_SUM` or `ISP_AF_ENV_USER_THRESHOLD_LUM` is non-zero, use the specified values as thresholds for sharpness and brightness changes
 - If `ISP_AF_ENV_USER_THRESHOLD_SUM` or `ISP_AF_ENV_USER_THRESHOLD_LUM` is zero, use the sharpness and brightness of the most recent statistics multiplied by `ISP_AF_ENV_THRESHOLD` (4 fractional bits) as the change threshold
4. Configure the statistical trigger mode. After each trigger completes, update the monitoring baseline;
 - If `ISP_AF_AUTO_UPDATE` is 1, it is in automatic triggering mode, which performs statistics for every frame with AF enabled. Since each statistical calculation clears the frame count for monitoring, the monitoring will be disabled
 - If `ISP_AF_AUTO_UPDATE` is 0, it is in manual triggering mode. The monitoring is working only under this condition
5. Enable AF by writing 1 to `ISP_AF_EN`;
6. Obtain statistics;
 - Clear the completion status by writing 1 to `ISP_AF_FDONE_INT_CLR`
 - Trigger statistics
 - With automatic triggering, statistics are performed for every frame
 - With manual triggering, write 1 to `ISP_AF_MANUAL_UPDATE` to trigger statistics manually
 - Wait for the statistics to complete
 - Wait for the ISP interrupt triggered by `AF_FDONE_INT`.

- Or query the status using [ISP_AF_FDONE_INT_RAW](#)
 - Obtain sharpness statistics by reading [ISP_AF_SUMA/B/C](#) and obtain brightness statistics by reading [ISP_AF_LUMA/B/C](#)
7. Performs focus scene monitoring.
- Wait for the ISP interrupt triggered by [AF_ENV_INT](#)
 - Or query the status using [ISP_AF_ENV_INT_RAW](#)
 - After detecting a change in the focus scene, disable monitoring, refocus the camera, and then re-enable monitoring

36.711 AWB Configuration

AWB can be configured after the ISP is enabled.

1. Select sampling points via [ISP_AWB_SAMPLE](#) and write 1 to [ISP_AWB_MODE](#);
2. Set up statistical windows;
 - Set the left boundary using [ISP_AWB_LPOINT](#)
 - Set the right boundary using [ISP_AWB_RPOINT](#)
 - Set the top boundary using [ISP_AWB_TPOINT](#)
 - Set the bottom boundary using [ISP_AWB_BPOINT](#)
3. Configure white patch constraints;
 - Set maximum luminance with [ISP_AWB_MAX_LUM](#)
 - Set minimum luminance with [ISP_AWB_MIN_LUM](#)
 - Set maximum R/G ratio with [ISP_AWB_MAX_RG](#) (2 integer bits, 8 fractional bits)
 - Set minimum R/G ratio with [ISP_AWB_MIN_RG](#) (2 integer bits, 8 fractional bits)
 - Set maximum B/G ratio with [ISP_AWB_MAX_BG](#) (2 integer bits, 8 fractional bits)
 - Set minimum B/G ratio with [ISP_AWB_MIN_BG](#) (2 integer bits, 8 fractional bits)
4. Obtain statistics;
 - Clear the completion status by writing 1 to [ISP_AWB_FDONE_INT_CLR](#)
 - Trigger statistics by writing 0 and then 1 to [ISP_AWB_EN](#)
 - Wait for the statistics to complete
 - Wait for the ISP interrupt triggered by [AWB_FDONE_INT](#)
 - Or query the status using [ISP_AWB_FDONE_INT_RAW](#)
 - Get the number of white patches within the statistical window by reading [ISP_AWBO_WHITE_CNT](#)
 - Get the accumulated values of the R, G, B components for the white patches within the statistical window by reading [ISP_AWBO_ACC_R/G/B](#)

36.7.12 HIST Configuration

HIST can be configured after the ISP is enabled.

1. Select sampling points via [ISP_HIST_MODE](#);
 - If sampling points are set to RGB, use [ISP_HIST_COEFF_R/G/B](#) to set RGB weights with 8 fractional bits
2. Set up statistical windows;
 - Set the horizontal start point with [ISP_HIST_X_OFFS](#)
 - Configure the horizontal dimension of each sub-window with [ISP_HIST_X_SIZE](#)
 - Set the vertical start point with [ISP_HIST_Y_OFFS](#)
 - Configure the vertical dimension of each sub-window with [ISP_HIST_Y_SIZE](#)
3. Configure histograms;
 - Set histogram interval divisions using [ISP_HIST_SEG_x_x](#)
 - Set window weights using [ISP_HIST_WEIGHT_xx](#) with 8 fractional bits
4. Enable HIST by writing 1 to [ISP_HIST_EN](#);
5. Obtain statistics.
 - Clear the completion status by writing 1 to [ISP_HIST_FDONE_INT_CLR](#)
 - With HIST enabled, statistics are performed at the end of every frame. Wait for the statistics to complete
 - Wait for the ISP interrupt triggered by [HIST_FDONE_INT](#)
 - Or query the status using [ISP_HIST_FDONE_INT_RAW](#).
 - Obtain the statistics for each interval by reading [ISP_HIST_BIN_x](#).

36.7.13 ISP_Pipeline Module Update Configuration

1. Disable the module via [ISP_CNTL_REG](#);
2. Wait for the interrupt from the corresponding module;
3. Update module configurations;
4. Re-enable the module.

36.8 Register Summary

36.8.1 ISP Register Summary

The addresses in this section are relative to **ISP** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Registers			
ISP_CLK_EN_REG	ISP clock control register	0x0004	R/W
ISP_CNTL_REG	ISP enable register	0x0008	R/W
ISP_HSYNC_CNT_REG	Hsync interval control register	0x000C	R/W
ISP_FRAME_CFG_REG	Image control register	0x0010	R/W
ISP_CCM_COEF0_REG	CCM parameter register 0	0x0014	R/W
ISP_CCM_COEF1_REG	CCM parameter register 1	0x0018	R/W
ISP_CCM_COEF3_REG	CCM parameter register 3	0x001C	R/W
ISP_CCM_COEF4_REG	CCM parameter register 4	0x0020	R/W
ISP_CCM_COEF5_REG	CCM parameter register 5	0x0024	R/W
ISP_BF_MATRIX_CTRL_REG	BF pixel-to-matrix control register	0x0028	R/W
ISP_BF_SIGMA_REG	BF noise reduction strength control register	0x002C	R/W
ISP_BF_GAU0_REG	BF noise reduction template register 0	0x0030	R/W
ISP_BF_GAU1_REG	BF noise reduction template register 1	0x0034	R/W
ISP_DPC_CTRL_REG	DPC control register	0x0038	R/W
ISP_DPC_CONF_REG	DPC parameter register	0x003C	R/W
ISP_DPC_MATRIX_CTRL_REG	DPC pixel-to-matrix control register	0x0040	R/W
ISP_LUT_CMD_REG	LUT command register	0x0048	WT
ISP_LUT_WDATA_REG	LUT write register	0x004C	R/W
ISP_LSC_TABLESIZE_REG	LSC parameter control register	0x0054	R/W
ISP_DEMOSAIC_MATRIX_CTRL_REG	Demosaic pixel-to-matrix control register	0x0058	R/W
ISP_DEMOSAIC_GRAD_RATIO_REG	Demosaic parameter control register	0x005C	R/W
ISP_GAMMA_CTRL_REG	Gamma correction control register	0x0074	R/W
ISP_GAMMA_RY1_REG	Gamma curve R channel Y axis configuration register 1	0x0078	R/W
ISP_GAMMA_RY2_REG	Gamma curve R channel Y axis configuration register 2	0x007C	R/W
ISP_GAMMA_RY3_REG	Gamma curve R channel Y axis configuration register 3	0x0080	R/W
ISP_GAMMA_RY4_REG	Gamma curve R channel Y axis configuration register 4	0x0084	R/W
ISP_GAMMA_GY1_REG	Gamma curve G channel Y axis configuration register 1	0x0088	R/W
ISP_GAMMA_GY2_REG	Gamma curve G channel Y axis configuration register 2	0x008C	R/W

Name	Description	Address	Access
ISP_GAMMA_GY3_REG	Gamma curve G channel Y axis configuration register 3	0x0090	R/W
ISP_GAMMA_GY4_REG	Gamma curve G channel Y axis configuration register 4	0x0094	R/W
ISP_GAMMA_BY1_REG	Gamma curve B channel Y axis configuration register 1	0x0098	R/W
ISP_GAMMA_BY2_REG	Gamma curve B channel Y axis configuration register 2	0x009C	R/W
ISP_GAMMA_BY3_REG	Gamma curve B channel Y axis configuration register 3	0x00A0	R/W
ISP_GAMMA_BY4_REG	Gamma curve B channel Y axis configuration register 4	0x00A4	R/W
ISP_GAMMA_RX1_REG	Gamma curve R channel X axis configuration register 1	0x00A8	R/W
ISP_GAMMA_RX2_REG	Gamma curve R channel X axis configuration register 2	0x00AC	R/W
ISP_GAMMA_GX1_REG	Gamma curve G channel X axis configuration register 1	0x00B0	R/W
ISP_GAMMA_GX2_REG	Gamma curve G channel X axis configuration register 2	0x00B4	R/W
ISP_GAMMA_BX1_REG	Gamma curve B channel X axis configuration register 1	0x00B8	R/W
ISP_GAMMA_BX2_REG	Gamma curve B channel X axis configuration register 2	0x00BC	R/W
ISP_AE_CTRL_REG	AE control register	0x00C0	varies
ISP_AE_MONITOR_REG	AE monitoring control register	0x00C4	R/W
ISP_AE_BX_REG	AE statistical window X direction configuration register	0x00C8	R/W
ISP_AE_BY_REG	AE statistical window Y direction configuration register	0x00CC	R/W
ISP_AE_WINPIXNUM_REG	AE sub-window pixel number configuration register	0x00D0	R/W
ISP_AE_WIN_RECIPROCAL_REG	AE sub-window pixel number reciprocal configuration register	0x00D4	R/W
ISP_SHARP_CTRL0_REG	Sharpen control register 0	0x00F4	R/W
ISP_SHARP_FILTER0_REG	Sharpen low-frequency template configuration register 0	0x00F8	R/W
ISP_SHARP_FILTER1_REG	Sharpen low-frequency template configuration register 1	0x00FC	R/W
ISP_SHARP_FILTER2_REG	Sharpen low-frequency template configuration register 2	0x0100	R/W
ISP_SHARP_MATRIX_CTRL_REG	Sharpen pixel-to-matrix control register	0x0104	R/W
ISP_SHARP_CTRL1_REG	Sharpen control register 1	0x0108	RO
ISP_DMA_CNTL_REG	ISP VDMA input control register	0x010C	varies

Name	Description	Address	Access
ISP_DMA_RAW_DATA_REG	ISP VDMA input data amount configuration register	0x0110	varies
ISP_CAM_CNTL_REG	ISP DVP input control register	0x0114	R/W
ISP_CAM_CONF_REG	ISP DVP input configuration register	0x0118	R/W
ISP_AF_CTRL0_REG	AF control register 0	0x011C	varies
ISP_AF_ENV_USER_TH_SUM_REG	AF scene monitoring sharpness threshold configuration register	0x0128	R/W
ISP_AF_ENV_USER_TH_LUM_REG	AF scene monitoring luminance threshold configuration register	0x012C	R/W
ISP_AF_THRESHOLD_REG	AF statistics threshold configuration register	0x0130	varies
ISP_AF_HSCALE_A_REG	AF statistical window A horizontal direction configuration register	0x0134	R/W
ISP_AF_VSCALE_A_REG	AF statistical window A vertical direction configuration register	0x0138	R/W
ISP_AF_HSCALE_B_REG	AF statistical window B horizontal direction configuration register	0x013C	R/W
ISP_AF_VSCALE_B_REG	AF statistical window B vertical direction configuration register	0x0140	R/W
ISP_AF_HSCALE_C_REG	AF statistical window C horizontal direction configuration register	0x0144	R/W
ISP_AF_VSCALE_C_REG	AF statistical window C vertical direction configuration register	0x0148	R/W
ISP_AWB_MODE_REG	AWB control register	0x0164	R/W
ISP_AWB_HSCALE_REG	AWB statistical window horizontal direction configuration register	0x0168	R/W
ISP_AWB_VSCALE_REG	AWB statistical window vertical direction configuration register	0x016C	R/W
ISP_AWB_TH_LUM_REG	AWB white patch luminance threshold configuration register	0x0170	R/W
ISP_AWB_TH_RG_REG	AWB white patch R/G threshold configuration register	0x0174	R/W
ISP_AWB_TH_BG_REG	AWB white patch B/G threshold configuration register	0x0178	R/W
ISP_COLOR_CTRL_REG	Contrast/saturation/luminance control register	0x018C	R/W
ISP_BLC_VALUE_REG	BLC control register	0x0190	R/W
ISP_BLC_CTRL0_REG	BLC pixel stretch register	0x0194	R/W
ISP_HIST_MODE_REG	HIST control register	0x01A4	R/W
ISP_HIST_COEFF_REG	HIST RGB parameter configuration register	0x01A8	R/W
ISP_HIST_OFFS_REG	HIST statistical window starting point configuration register	0x01AC	R/W
ISP_HIST_SIZE_REG	HIST statistical sub-window size configuration register	0x01B0	R/W
ISP_HIST_SEGO_REG	HIST X coordinate interval configuration register 0	0x01B4	R/W

Name	Description	Address	Access
ISP_HIST_SEG1_REG	HIST X coordinate interval configuration register 1	0x01B8	R/W
ISP_HIST_SEG2_REG	HIST X coordinate interval configuration register 2	0x01BC	R/W
ISP_HIST_SEG3_REG	HIST X coordinate interval configuration register 3	0x01C0	R/W
ISP_HIST_WEIGHT0_REG	HIST sub-window weight configuration register 0	0x01C4	R/W
ISP_HIST_WEIGHT1_REG	HIST sub-window weight configuration register 1	0x01C8	R/W
ISP_HIST_WEIGHT2_REG	HIST sub-window weight configuration register 2	0x01CC	R/W
ISP_HIST_WEIGHT3_REG	HIST sub-window weight configuration register 3	0x01D0	R/W
ISP_HIST_WEIGHT4_REG	HIST sub-window weight configuration register 4	0x01D4	R/W
ISP_HIST_WEIGHT5_REG	HIST sub-window weight configuration register 5	0x01D8	R/W
ISP_HIST_WEIGHT6_REG	HIST sub-window weight configuration register 6	0x01DC	R/W
ISP_YUV_FORMAT_REG	YUV format control register	0x0234	R/W
ISP_CROP_CTRL_REG	CROP control register	0x0244	WT
ISP_CROP_Y_CAPTURE_REG	CROP Y direction control register	0x0248	R/W
ISP_CROP_X_CAPTURE_REG	CROP X direction control register	0x024C	R/W
ISP_CROP_ERR_ST_REG	CROP error status register	0x0250	RO
ISP_WBG_COEF_R_REG	WBG R channel gain register	0x0254	R/W
ISP_WBG_COEF_G_REG	WBG G channel gain register	0x0258	R/W
ISP_WBG_COEF_B_REG	WBG B channel gain register	0x025C	R/W
ISP_COLOR_HUE_CTRL_REG	Hue control register	0x0260	R/W
ISP_AWB_BX_REG	AWB sub-window X direction control register	0x0264	R/W
ISP_AWB_BY_REG	AWB sub-window Y direction control register	0x0268	R/W
ISP_STATE_REG	ISP state register	0x026C	RO
ISP_SHADOW_REG_CTRL_REG	Shadow register control register	0x0270	R/W
Status Registers			
ISP_DPC_DEADPIX_CNT_REG	DPC static calibration defective pixel number register	0x0044	RO
ISP_LUT_RDATA_REG	LUT read register	0x0050	RO
ISP_AE_BLOCK_MEAN_0_REG	AE statistics register 0	0x00D8	RO
ISP_AE_BLOCK_MEAN_1_REG	AE statistics register 1	0x00DC	RO
ISP_AE_BLOCK_MEAN_2_REG	AE statistics register 2	0x00E0	RO
ISP_AE_BLOCK_MEAN_3_REG	AE statistics register 3	0x00E4	RO
ISP_AE_BLOCK_MEAN_4_REG	AE statistics register 4	0x00E8	RO
ISP_AE_BLOCK_MEAN_5_REG	AE statistics register 5	0x00EC	RO
ISP_AE_BLOCK_MEAN_6_REG	AE statistics register 6	0x00F0	RO

Name	Description	Address	Access
ISP_AF_SUM_A_REG	AF window A sharpness statistics register	0x014C	RO
ISP_AF_SUM_B_REG	AF window B sharpness statistics register	0x0150	RO
ISP_AF_SUM_C_REG	AF window C sharpness statistics register	0x0154	RO
ISP_AF_LUM_A_REG	AF window A luminance statistics register	0x0158	RO
ISP_AF_LUM_B_REG	AF window B luminance statistics register	0x015C	RO
ISP_AF_LUM_C_REG	AF window C luminance statistics register	0x0160	RO
ISP_AWBO_WHITE_CNT_REG	AWB white patch count statistics register	0x017C	RO
ISP_AWBO_ACC_R_REG	AWB R channel statistics register	0x0180	RO
ISP_AWBO_ACC_G_REG	AWB G channel statistics register	0x0184	RO
ISP_AWBO_ACC_B_REG	AWB B channel statistics register	0x0188	RO
ISP_HIST_BINO_REG	HIST interval 0 statistics register	0x01E0	RO
ISP_HIST_BIN1_REG	HIST interval 1 statistics register	0x01E4	RO
ISP_HIST_BIN2_REG	HIST interval 2 statistics register	0x01E8	RO
ISP_HIST_BIN3_REG	HIST interval 3 statistics register	0x01EC	RO
ISP_HIST_BIN4_REG	HIST interval 4 statistics register	0x01F0	RO
ISP_HIST_BIN5_REG	HIST interval 5 statistics register	0x01F4	RO
ISP_HIST_BIN6_REG	HIST interval 6 statistics register	0x01F8	RO
ISP_HIST_BIN7_REG	HIST interval 7 statistics register	0x01FC	RO
ISP_HIST_BIN8_REG	HIST interval 8 statistics register	0x0200	RO
ISP_HIST_BIN9_REG	HIST interval 9 statistics register	0x0204	RO
ISP_HIST_BIN10_REG	HIST interval 10 statistics register	0x0208	RO
ISP_HIST_BIN11_REG	HIST interval 11 statistics register	0x020C	RO
ISP_HIST_BIN12_REG	HIST interval 12 statistics register	0x0210	RO
ISP_HIST_BIN13_REG	HIST interval 13 statistics register	0x0214	RO
ISP_HIST_BIN14_REG	HIST interval 14 statistics register	0x0218	RO
ISP_HIST_BIN15_REG	HIST interval 15 statistics register	0x021C	RO
Interrupt Registers			
ISP_INT_RAW_REG	Raw interrupt status register	0x0064	R/SS/ WTC
ISP_INT_ST_REG	Masked interrupt status register	0x0068	RO
ISP_INT_ENA_REG	Interrupt enable register	0x006C	R/W
ISP_INT_CLR_REG	Interrupt clear register	0x0070	WT
Version Register			
ISP_VER_DATE_REG	Version control register	0x0000	R/W

36.8.2 MIPI CSI_Bridge Register Summary

The addresses in this section are relative to **MIPI CSI_Bridge** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Clock Gating Control Register			

Name	Description	Address	Access
CSI_BRIG_CLK_EN_REG	CSI_Bridge clock gate register	0x0000	R/W
CSI_Bridge Control Registers			
CSI_BRIG_CSI_EN_REG	CSI_Bridge enable register	0x0004	R/W
CSI_BRIG_BUF_FLOW_CTL_REG	CSI_Bridge buffer control register	0x000C	varies
DMA Control Registers			
CSI_BRIG_DMA_REQ_CFG_REG	DMA request control register	0x0008	R/W
CSI_BRIG_DMA_REQ_INTERVAL_REG	DMA request interval register	0x002C	R/W
CSI_BRIG_DMABLK_SIZE_REG	DMA data block size control register	0x0030	R/W
Frame Format Configuration Registers			
CSI_BRIG_DATA_TYPE_CFG_REG	DATA_TYPE control register	0x0010	R/W
CSI_BRIG_FRAME_CFG_REG	Image control register	0x0014	R/W
CSI_BRIG_ENDIAN_MODE_REG	Byte order control register	0x0018	R/W
Interrupt Registers			
CSI_BRIG_INT_RAW_REG	CSI_Bridge raw interrupt status register	0x001C	R/ WTC/ SS
CSI_BRIG_INT_CLR_REG	CSI_Bridge interrupt clear register	0x0020	WT
CSI_BRIG_INT_ST_REG	CSI_Bridge masked interrupt status register	0x0024	RO
CSI_BRIG_INT_ENA_REG	CSI_Bridge interrupt enable register	0x0028	R/W
CSI HOST Control Register			
CSI_BRIG_HOST_CTRL_REG	CSI HOST control register	0x0040	R/W

36.9 Registers

36.9.1 ISP Registers

The addresses in this section are relative to **ISP** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 36.1. ISP_VER_DATE_REG (0x0000)

ISP_VER_DATA	
31	0
0x20210608	
Reset	

ISP_VER_DATA ISP version control register. (R/W)

Register 36.2. ISP_CLK_EN_REG (0x0004)

(reserved)												ISP_CLK_WBG_FORCE_ON ISP_CLK_CROP_FORCE_ON ISP_MEM_CLK_FORCE_ON ISP_CLK_MIP1_ID1_FORCE_ON ISP_CLK_HIST_FORCE_ON ISP_CLK_AWB_FORCE_ON ISP_CLK_AF_FORCE_ON ISP_CLK_AE_FORCE_ON ISP_CLK_YUV2RGB_FORCE_ON ISP_CLK_COLOR_FORCE_ON ISP_CLK_SHARP_FORCE_ON ISP_CLK_RGB2YUV_FORCE_ON (reserved) ISP_CLK_GAMMA_FORCE_ON ISP_CLK_CCM_FORCE_ON ISP_CLK_DEMOAIC_FORCE_ON ISP_CLK_LSC_FORCE_ON ISP_CLK_BF_FORCE_ON ISP_CLK_DPC_FORCE_ON ISP_CLK_BLC_FORCE_ON ISP_CLK_EN																					
31												21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

ISP_CLK_EN Configures whether the ISP clock is always on.

0: The clock is enabled only during register access

1: The clock is always on

(R/W)

ISP_CLK_BLC_FORCE_ON Configures whether to force enable the BLC clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_DPC_FORCE_ON Configures whether to force enable the DPC clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_BF_FORCE_ON Configures whether to force enable the BF clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_LSC_FORCE_ON Configures whether to force enable the LSC clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_DEMOAIC_FORCE_ON Configures whether to force enable the demosaic clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_CCM_FORCE_ON Configures whether to force enable the CCM clock.

0: Not force enable

1: Force enable

(R/W)

Continued on the next page...

Register 36.2. ISP_CLK_EN_REG (0x0004)

Continued from the previous page...

ISP_CLK_GAMMA_FORCE_ON Configures whether to force enable the gamma correction clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_RGB2YUV_FORCE_ON Configures whether to force enable the RGB2YUV clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_SHARP_FORCE_ON Configures whether to force enable the sharpen clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_COLOR_FORCE_ON Configures whether to force enable the contrast/hue/saturation/luminance clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_YUV2RGB_FORCE_ON Configures whether to force enable the YUV2RGB clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_AE_FORCE_ON Configures whether to force enable the AE clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_AF_FORCE_ON Configures whether to force enable the AF clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_AWB_FORCE_ON Configures whether to force enable the AWB clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_HIST_FORCE_ON Configures whether to force enable the HIST clock.

0: Not force enable

1: Force enable

(R/W)

Continued on the next page...

Register 36.2. ISP_CLK_EN_REG (0x0004)

Continued from the previous page...

ISP_CLK_MIPI_IDI_FORCE_ON Configures whether to force enable the MIPI Image Interface 32 input clock.

0: Not force enable

1: Force enable

(R/W)

ISP_MEM_CLK_FORCE_ON Configures whether to force enable all the ISP memory clocks.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_CROP_FORCE_ON Configures whether to force enable the CROP clock.

0: Not force enable

1: Force enable

(R/W)

ISP_CLK_WBG_FORCE_ON Configures whether to force enable the WBG clock.

0: Not force enable

1: Force enable

(R/W)

Register 36.3. ISP_CNTL_REG (0x0008)

ISP_OUT_TYPE		ISP_IN_SRC		ISP_DATA_TYPE		ISP_BYTE_ENDIAN_ORDER		(reserved)		ISP_WBG_EN		ISP_CROP_EN		ISP_HIST_EN		ISP_AWB_EN		ISP_AF_EN		ISP_AE_EN		ISP_YUV2RGB_EN		ISP_COLOR_EN		ISP_SHARP_EN		ISP_RGB2YUV_EN		ISP_GAMMA_EN		ISP_CCM_EN		(reserved)		ISP_DEMOSAIC_EN		ISP_LSC_EN		ISP_BF_EN		ISP_DPC_EN		ISP_BLC_EN		ISP_MIPI_DATA_EN										
31	29	28	27	26	25	24	23	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																												
0x2		0		0		0		0 0 0 0		0		0		0		0		0		1		0		0		1		0		0		0		1		0		0		0		0		1		0		Reset								

ISP_MIPI_DATA_EN Configures whether to enable the MIPI Image Interface 32 input.

0: Disable

1: Enable

(R/W)

ISP_EN Configures whether to enable the ISP.

0: Disable

1: Enable

(R/W)

ISP_BLC_EN Configures whether to enable the BLC module.

0: Disable

1: Enable

(R/W)

ISP_DPC_EN Configures whether to enable the DPC module.

0: Disable

1: Enable

(R/W)

ISP_BF_EN Configures whether to enable the BF module.

0: Disable

1: Enable

(R/W)

ISP_LSC_EN Configures whether to enable the LSC module.

0: Disable

1: Enable

(R/W)

ISP_DEMOSAIC_EN Configures whether to enable the demosaic module.

0: Disable

1: Enable

(R/W)

ISP_CCM_EN Configures whether to enable the CCM module.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 36.3. ISP_CNTL_REG (0x0008)

Continued from the previous page...

ISP_GAMMA_EN Configures whether to enable the gamma correction module.

0: Disable

1: Enable

(R/W)

ISP_RGB2YUV_EN Configures whether to enable the RGB2YUV module.

0: Disable

1: Enable

(R/W)

ISP_SHARP_EN Configures whether to enable the sharpen module.

0: Disable

1: Enable

(R/W)

ISP_COLOR_EN Configures whether to enable the contrast/hue/saturation/luminance module.

0: Disable

1: Enable

(R/W)

ISP_YUV2RGB_EN Configures whether to enable the YUV2RGB module.

0: Disable

1: Enable

(R/W)

ISP_AE_EN Configures whether to enable the AE module.

0: Disable

1: Enable

(R/W)

ISP_AF_EN Configures whether to enable the AF module.

0: Disable

1: Enable

(R/W)

ISP_AWB_EN Configures whether to enable the AWB module.

0: Disable

1: Enable

(R/W)

ISP_HIST_EN Configures whether to enable the HIST module.

0: Disable

1: Enable

(R/W)

Continued on the next page...

Register 36.3. ISP_CNTL_REG (0x0008)

Continued from the previous page...

ISP_CROP_EN Configures whether to enable the CROP module.

0: Disable

1: Enable

(R/W)

ISP_WBG_EN Configures whether to enable the WBG module.

0: Disable

1: Enable

(R/W)

ISP_BYTE_ENDIAN_ORDER Configures the byte order for MIPI Image Interface 32 input when ISP is disabled.

0: csi_data[31:0]

1: [7:0], [15:8], [23:16], [31:24]

(R/W)

ISP_DATA_TYPE Configures the input data type.

0: RAW8

1: RAW10

2: RAW12

3: invalid

(R/W)

ISP_IN_SRC Configures the input data source.

0: MIPI Image Interface 32 data from CSI HOST

1: Data from DVP

2: Data from VDMA

3: invalid

(R/W)

ISP_OUT_TYPE Configures the output data type.

0: RAW8

1: YUV422

2: RGB888

3: YUV420

4: RGB565

Others: Invalid

(R/W)

Register 36.4. ISP_HSYNC_CNT_REG (0x000C)

(reserved)																ISP_HSYNC_CNT																	
31																8	7	0															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	Reset						

ISP_HSYNC_CNT Configures the interval between hsync and the previous vsync or line_end when converting Image Interface 32 data to ISP data. Generally, the default value is used. Measurement unit: Number of ISP_CLK clock cycles. (R/W)

Register 36.5. ISP_FRAME_CFG_REG (0x0010)

(reserved)				ISP_HSYNC_END_EXIST				ISP_HSYNC_START_EXIST				ISP_BAYER_MODE				(reserved)				ISP_HADR_NUM												ISP_VADR_NUM													
31	30	29	28	27	26	24			23	12										11	0																								
0	1	1	0	0	0	0	480			480										Reset																									

ISP_VADR_NUM Configures the height of the input image, which should be set as the actual number of lines - 1. (R/W)

ISP_HADR_NUM Configures the width of the input image, which should be set as the actual width of lines - 1. (R/W)

ISP_BAYER_MODE Configures the Bayer mode of the input image.

0: BG/GR

1: GB/RG

2: GR/BG

3: RG/GB

(R/W)

ISP_HSYNC_START_EXIST Configures whether the hsync_start packet exists.

0: Not exist

1: Exist

(R/W)

ISP_HSYNC_END_EXIST Configures whether the hsync_end packet exists. This must have the same value as the [ISP_HSYNC_START_EXIST](#) field.

0: Not exist

1: Exist

(R/W)

Register 36.6. ISP_CCM_COEF0_REG (0x0014)

(reserved)						ISP_CCM_RG							ISP_CCM_RR																					
31						26											13	12																0
0	0	0	0	0	0	4736										1856															Reset			

ISP_CCM_RR Configures the CCM RR parameter. The 12th bit is the sign bit, and bits [11:0] represent the absolute value. Bits [11:10] are the integer part, and bits [9:0] are the fractional part. (R/W)

ISP_CCM_RG Configures the CCM RG parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

Register 36.7. ISP_CCM_COEF1_REG (0x0018)

(reserved)						ISP_CCM_GR							ISP_CCM_RB															
31		26		25								13		12														0
0	0	0	0	0	0	0																						Reset

ISP_CCM_RB Configures the CCM RB parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

ISP_CCM_GR Configures the CCM GR parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

Register 36.8. ISP_CCM_COEF3_REG (0x001C)

(reserved)						ISP_CCM_GB							ISP_CCM_GG															
31		26		25								13		12														0
0	0	0	0	0	0	0																						Reset

ISP_CCM_GG Configures the CCM GG parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

ISP_CCM_GB Configures the CCM GB parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

Register 36.9. ISP_CCM_COEF4_REG (0x0020)

(reserved)						ISP_CCM_BG						ISP_CCM_BR						
31	26					25	13					12	0					
0	0	0	0	0	0	4800						4160						Reset

- ISP_CCM_BR Configures the CCM BR parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)
- ISP_CCM_BG Configures the CCM BG parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

Register 36.10. ISP_CCM_COEF5_REG (0x0024)

(reserved)																ISP_CCM_BB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				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- ISP_CCM_BB Configures the CCM BB parameter, following the same bit structure and interpretation as [ISP_CCM_RR](#). (R/W)

Register 36.11. ISP_BF_MATRIX_CTRL_REG (0x0028)

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ISP_BF_TAIL_PIXEN_PULSE_TL Configures the data control cycle for the tail-row data rate during pixel-to-matrix conversion in the BF module.

Within [ISP_BF_TAIL_PIXEN_PULSE_TL](#) clock cycles, only the first [ISP_BF_TAIL_PIXEN_PULSE_TH](#) cycles are valid. Be cautious to configure this field, to prevent data processing from being deferred to the next frame.

This feature is enabled only when both [ISP_BF_TAIL_PIXEN_PULSE_TL](#) and [ISP_BF_TAIL_PIXEN_PULSE_TH](#) are non-zero, and [ISP_BF_TAIL_PIXEN_PULSE_TH](#) < [ISP_BF_TAIL_PIXEN_PULSE_TL](#). (R/W)

ISP_BF_TAIL_PIXEN_PULSE_TH Configures the valid data cycle of the tail-row data rate during pixel-to-matrix conversion in the BF module, which must be less than [ISP_HADR_NUM](#) - 1. (R/W)

ISP_BF_PADDING_DATA Configures the padding data used during image edge expansion in pixel-to-matrix conversion of the BF module. (R/W)

ISP_BF_PADDING_MODE Configures the method of image edge expansion during pixel-to-matrix conversion in the BF module.

0: Automatically padded with pixel values from the image edge

1: Padded with the data in [ISP_BF_PADDING_DATA](#)

(R/W)

Register 36.12. ISP_BF_SIGMA_REG (0x002C)

(reserved)																								ISP - SIGMA									
31																								6	5	0							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	Reset					

ISP_SIGMA Configures the strength of noise reduction in the BF module. Values from 2 to 20 represent increasing strength, while other values indicate maximum strength. (R/W)

Register 36.13. ISP_BF_GAU0_REG (0x0030)

[illegible]

ISP_GAU_TEMPLATE21	Configures the value of coordinate 21 in the noise reduction template of the BF module. (R/W)
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ISP_GAU_TEMPLATE20 Configures the value of coordinate 20 in the noise reduction template of the BF module. (R/W)

ISP_GAU_TEMPLATE12	Configures the value of coordinate 12 in the noise reduction template of the BF module. (R/W)
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ISP_GAU_TEMPLATE11	Configures the value of coordinate 11 in the noise reduction template of the BF module. (R/W)
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ISP_GAU_TEMPLATE10	Configures the value of coordinate 10 in the noise reduction template of the BF module. (R/W)
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ISP_GAU_TEMPLATE02 Configures the value of coordinate 02 in the noise reduction template of the BF module. (R/W)

ISP_GAU_TEMPLATE01	Configures the value of coordinate 01 in the noise reduction template of the BF module. (R/W)
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ISP_GAU_TEMPLATE00 Configures the value of coordinate 00 in the noise reduction template of the BF module. (R/W)

Register 36.14. ISP_BF_GAU1_REG (0x0034)

Diagram illustrating the structure of the `ISP_GAU_TEMPLATE22` register. The register is 32 bits wide, divided into two main sections:

- Reserved Field (Bits 31-4):** Labeled `(reserved)`, occupying 28 bits.
- ISP_GAU_TEMPLATE22 Field (Bits 3-0):** Labeled `ISP_GAU_TEMPLATE22`, occupying 4 bits. The value `0xf` is shown in this field, and the label `Reset` is placed below it.

ISP_GAU_TEMPLATE22 Configures the value of coordinate 22 in the noise reduction template of the BF module. (R/W)

Register 36.15. ISP_DPC_CTRL_REG (0x0038)

(reserved)																										ISP_DPC_CHECK_OD_EN						ISP_DPC_METHOD_SEL						ISP_DPC_BLACK_EN						ISP_DYN_EN						ISP_STA_EN						ISP_DPC_CHECK_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													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Reset

ISP_DPC_CHECK_EN Configures whether to enable the static calibration function. Calibration and correction cannot be enabled at the same time.

0: Disable

1: Enable

(R/W)

ISP_STA_EN Configures whether to enable the static correction function.

0: Disable

1: Enable

(R/W)

ISP_DYN_EN Configures whether to enable the dynamic correction function.

0: Disable

1: Enable

(R/W)

ISP_DPC_BLACK_EN Configures the image type used for static calibration.

0: White image

1: Black image

(R/W)

ISP_DPC_METHOD_SEL Configures the algorithm selection for dynamic correction.

0: Select Algorithm 0

1: Select Algorithm 1

(R/W)

ISP_DPC_CHECK_OD_EN Configures whether to output image data during the static calibration process.

0: Do not output image

1: Output image

(R/W)

Register 36.16. ISP_DPC_CONF_REG (0x003C)

(reserved)				ISP_DPC_FACTOR_BRIG				ISP_DPC_FACTOR_DARK				ISP_DPC_THRESHOLD_H				ISP_DPC_THRESHOLD_L			
31	28	27	22	21	16	15	8	7											
0	0	0	0	0x10				0x10				0x30				0x30			

Reset

ISP_DPC_THRESHOLD_L Configures the defective pixel threshold for black image static calibration, or the low threshold in dynamic correction Algorithm 0 (8-bit valid), or the low threshold in dynamic correction Algorithm 1 (5-bit valid, maximum value is 16). For details, refer to Section [36.5.2.2](#). (R/W)

ISP_DPC_THRESHOLD_H Configures the defective pixel threshold for white image static calibration, or the high threshold in dynamic correction Algorithm 0 (8-bit valid), or the high threshold in dynamic correction Algorithm 1 (5-bit valid, maximum value is 16). For details, refer to Section [36.5.2.2](#). (R/W)

ISP_DPC_FACTOR_DARK Configures the dark pixel factor used in dynamic correction Algorithm 1. (R/W)

ISP_DPC_FACTOR_BRIG Configures the bright pixel factor used in dynamic correction Algorithm 1. (R/W)

Register 36.17. ISP_DPC_MATRIX_CTRL_REG (0x0040)

(reserved)								ISP_DPC_PADDING_MODE								ISP_DPC_PADDING_DATA								ISP_DPC_TAIL_PIXEN_PULSE_TH								ISP_DPC_TAIL_PIXEN_PULSE_TL																																																															
31								25								24	23								16								15								8								7								0																																						
0								0								0								0								0								0								0								0								0x0								0x0								0x0								Reset							

Reset

ISP_DPC_TAIL_PIXEN_PULSE_TL Configures the data control cycle for the tail-row data rate during pixel-to-matrix conversion in the DPC module. Within the [ISP_DPC_TAIL_PIXEN_PULSE_TL](#) clock cycles, only the first [ISP_DPC_TAIL_PIXEN_PULSE_TH](#) cycles are valid. Be cautious to configure this field, to prevent data processing from being deferred to the next frame.

This function is enabled only when both [ISP_DPC_TAIL_PIXEN_PULSE_TL](#) and [ISP_DPC_TAIL_PIXEN_PULSE_TH](#) are non-zero, and [ISP_DPC_TAIL_PIXEN_PULSE_TH](#) < [ISP_DPC_TAIL_PIXEN_PULSE_TL](#). (R/W)

ISP_DPC_TAIL_PIXEN_PULSE_TH Configures the valid data cycle of the tail-row data rate during pixel-to-matrix conversion in the DPC module, which must be less than [ISP_HADR_NUM](#) - 1. (R/W)

ISP_DPC_PADDING_DATA Configures the padding data used for image edge expansion during pixel-to-matrix conversion in the DPC module. (R/W)

ISP_DPC_PADDING_MODE Configures the method of image edge expansion during pixel-to-matrix conversion in the DPC module.

0: Automatically padded with pixel values from the image edge

1: Padded with the data in [ISP_DPC_PADDING_DATA](#)

(R/W)

Register 36.18. ISP_LUT_CMD_REG (0x0048)

(reserved)																ISP_LUT_CMD		ISP_LUT_NUM				ISP_LUT_ADDR											
31																17		16	15	12				11								0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0		0x0		0x0								Reset					

ISP_LUT_ADDR Configures the LUT address. When [ISP_LUT_NUM](#) selects LSC LUT, [11:10] being 0 indicates selecting Gb_B LUT, and being 1 indicates selecting R_Gr LUT. Bits [9:0] represent the actual LUT address.

(WT)

ISP_LUT_NUM Configures the LUT selection.

0: Select LSC LUT

Others: Invalid

(WT)

ISP_LUT_CMD Configures whether to read from or write to the LUT.

0: Read from the LUT

1: Write to the LUT

(WT)

Register 36.19. ISP_LUT_WDATA_REG (0x004C)

ISP_LUT_WDATA																																
31																															0	
0x000000																																Reset

ISP_LUT_WDATA Data to be written to the LUT, which must be configured before writing to [ISP_LUT_CMD_REG](#). (R/W)

Register 36.20. ISP_LSC_TABLESIZE_REG (0x0054)

(reserved)																															ISP_LSC_XTABLESIZE									
31																															5	4	0							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	31	Reset									

ISP_LSC_XTABLESIZE Configures the number of horizontal grids in the LSC module. This value is obtained by $fix((line_number - 1)/2/32) + 2$, where $fix()$ denotes the rounding. (R/W)

Register 36.21. ISP_DEMOSAIC_MATRIX_CTRL_REG (0x0058)

(reserved)								ISP_DEMOSAIC_PADDING_MODE								ISP_DEMOSAIC_PADDING_DATA								ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TH								ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TL																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TL Configures the data control cycle for the tail-row data rate during pixel-to-matrix conversion in the demosaic module.

Within [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TL](#) clock cycles, only the first [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TH](#) cycles are valid. Be cautious to configure this field, to prevent data processing from being deferred to the next frame.

This feature is enabled only when both [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TL](#) and [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TH](#) are non-zero, and [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TH](#) < [ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TL](#). (R/W)

ISP_DEMOSAIC_TAIL_PIXEN_PULSE_TH Configures the valid data cycle of the tail-row data rate during pixel-to-matrix conversion in the demosaic module, which must be less than [ISP_HADR_NUM](#) - 1. (R/W)

ISP_DEMOSAIC_PADDING_DATA Configures the padding data used during image edge expansion in pixel-to-matrix conversion of the demosaic module. (R/W)

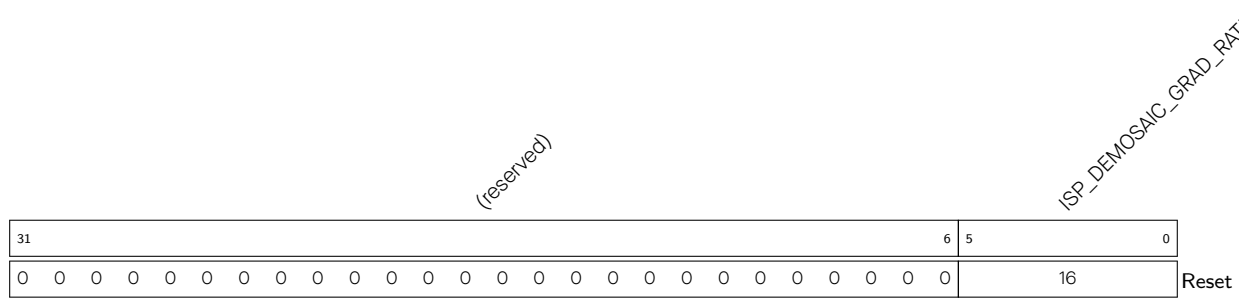
ISP_DEMOSAIC_PADDING_MODE Configures the method of image edge expansion during pixel-to-matrix conversion in the demosaic module.

0: Automatically padded with pixel values from the image edge

1: Padded with the data in [ISP_DEMOSAIC_PADDING_DATA](#)

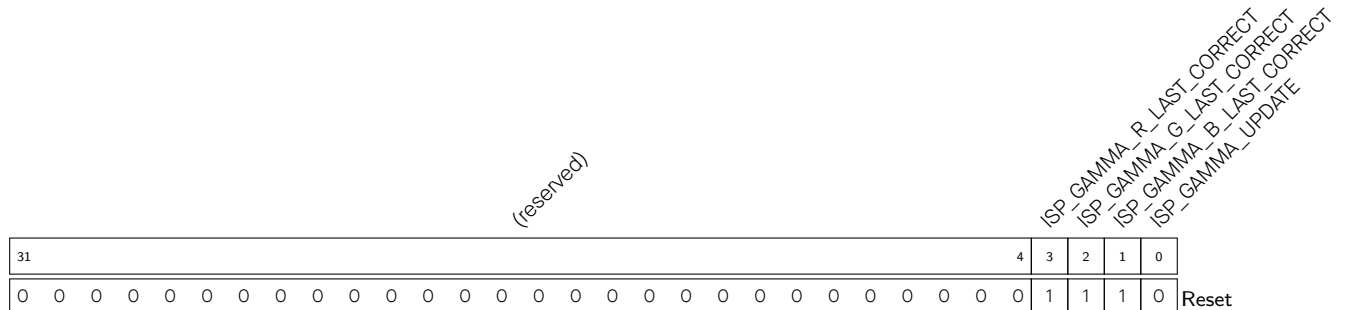
(R/W)

Register 36.22. ISP_DEMOSAIC_GRAD_RATIO_REG (0x005C)



ISP_DEMOSAIC_GRAD_RATIO Configures the demosaic effect, with bits [5:4] as the integer part and bits [3:0] as the fractional part. (R/W)

Register 36.23. ISP_GAMMA_CTRL_REG (0x0074)



ISP_GAMMA_UPDATE Configures whether to update the gamma curve parameters. Writing 1 to this field updates the gamma curve parameters. It will be automatically cleared to 0 after the update is complete. (R/W)

ISP_GAMMA_B_LAST_CORRECT Configures whether to enable correction for the parameters of the last interval on the X axis of B channel gamma curve.
 0: Not enable
 1: Enable
 (R/W)

ISP_GAMMA_G_LAST_CORRECT Configures whether to enable correction for the parameters of the last interval on the X axis of G channel gamma curve.
 0: Not enable
 1: Enable
 (R/W)

ISP_GAMMA_R_LAST_CORRECT Configures whether to enable correction for the parameters of the last interval on the X axis of X channel gamma curve.
 0: Not enable
 1: Enable
 (R/W)

Register 36.24. ISP_GAMMA_RY1_REG (0x0078)

ISP_GAMMA_R_Y00							
ISP_GAMMA_R_Y01							
ISP_GAMMA_R_Y02							
ISP_GAMMA_R_Y03							
31	24	23	16	15	8	7	0
0x10		0x20		0x30		0x40	
Reset							

ISP_GAMMA_R_Y03 Configures the value of the 3rd point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y02 Configures the value of the 2nd point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y01 Configures the value of the 1st point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y00 Configures the value of the 0th point on the Y axis of R channel gamma curve.
(R/W)

Register 36.25. ISP_GAMMA_RY2_REG (0x007C)

ISP_GAMMA_R_Y04							
ISP_GAMMA_R_Y05							
ISP_GAMMA_R_Y06							
ISP_GAMMA_R_Y07							
31	24	23	16	15	8	7	0
0x50		0x60		0x70		0x80	
Reset							

ISP_GAMMA_R_Y07 Configures the value of the 7th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y06 Configures the value of the 6th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y05 Configures the value of the 5th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y04 Configures the value of the 4th point on the Y axis of R channel gamma curve.
(R/W)

Register 36.26. ISP_GAMMA_RY3_REG (0x0080)

ISP_GAMMA_R_Y08							
ISP_GAMMA_R_Y09							
ISP_GAMMA_R_Y0A							
ISP_GAMMA_R_Y0B							
31	24	23	16	15	8	7	0
0x90							
0xa0							
0xb0							
0xc0							
Reset							

ISP_GAMMA_R_Y0B Configures the value of the 11th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y0A Configures the value of the 10th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y09 Configures the value of the 9th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y08 Configures the value of the 8th point on the Y axis of R channel gamma curve.
(R/W)

Register 36.27. ISP_GAMMA_RY4_REG (0x0084)

ISP_GAMMA_R_Y0C							
ISP_GAMMA_R_Y0D							
ISP_GAMMA_R_Y0E							
ISP_GAMMA_R_Y0F							
31	24	23	16	15	8	7	0
0xd0							
0xe0							
0xf0							
0xff							
Reset							

ISP_GAMMA_R_Y0F Configures the value of the 15th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y0E Configures the value of the 14th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y0D Configures the value of the 13th point on the Y axis of R channel gamma curve.
(R/W)

ISP_GAMMA_R_Y0C Configures the value of the 12th point on the Y axis of R channel gamma curve.
(R/W)

Register 36.28. ISP_GAMMA_GY1_REG (0x0088)

ISP_GAMMA_G_Y00							
ISP_GAMMA_G_Y01							
ISP_GAMMA_G_Y02							
ISP_GAMMA_G_Y03							
31	24	23	16	15	8	7	0
Reset							
0x10							
0x20							
0x30							
0x40							

ISP_GAMMA_G_Y03 Configures the value of the 3rd point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y02 Configures the value of the 2nd point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y01 Configures the value of the 1st point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y00 Configures the value of the 0th point on the Y axis of G channel gamma curve.
(R/W)

Register 36.29. ISP_GAMMA_GY2_REG (0x008C)

ISP_GAMMA_G_Y04							
ISP_GAMMA_G_Y05							
ISP_GAMMA_G_Y06							
ISP_GAMMA_G_Y07							
31	24	23	16	15	8	7	0
Reset							
0x50							
0x60							
0x70							
0x80							

ISP_GAMMA_G_Y07 Configures the value of the 7th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y06 Configures the value of the 6th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y05 Configures the value of the 5th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y04 Configures the value of the 4th point on the Y axis of G channel gamma curve.
(R/W)

Register 36.30. ISP_GAMMA_GY3_REG (0x0090)

ISP_GAMMA_G_Y08							
ISP_GAMMA_G_Y09							
ISP_GAMMA_G_Y0A							
ISP_GAMMA_G_Y0B							
31	24	23	16	15	8	7	0
0x90							
0xa0							
0xb0							
0xc0							
Reset							

ISP_GAMMA_G_Y0B Configures the value of the 11th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y0A Configures the value of the 10th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y09 Configures the value of the 9th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y08 Configures the value of the 8th point on the Y axis of G channel gamma curve.
(R/W)

Register 36.31. ISP_GAMMA_GY4_REG (0x0094)

ISP_GAMMA_G_Y0C							
ISP_GAMMA_G_Y0D							
ISP_GAMMA_G_Y0E							
ISP_GAMMA_G_Y0F							
31	24	23	16	15	8	7	0
0xd0							
0xe0							
0xf0							
0xff							
Reset							

ISP_GAMMA_G_Y0F Configures the value of the 15th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y0E Configures the value of the 14th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y0D Configures the value of the 13th point on the Y axis of G channel gamma curve.
(R/W)

ISP_GAMMA_G_Y0C Configures the value of the 12th point on the Y axis of G channel gamma curve.
(R/W)

Register 36.32. ISP_GAMMA_BY1_REG (0x0098)

ISP_GAMMA_B_Y00							
ISP_GAMMA_B_Y01							
ISP_GAMMA_B_Y02							
ISP_GAMMA_B_Y03							
31	24	23	16	15	8	7	0
0x10							
0x20							
0x30							
0x40							
Reset							

ISP_GAMMA_B_Y03 Configures the value of the 3rd point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y02 Configures the value of the 2nd point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y01 Configures the value of the 1st point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y00 Configures the value of the 0th point on the Y axis of B channel gamma curve.
(R/W)

Register 36.33. ISP_GAMMA_BY2_REG (0x009C)

ISP_GAMMA_B_Y04							
ISP_GAMMA_B_Y05							
ISP_GAMMA_B_Y06							
ISP_GAMMA_B_Y07							
31	24	23	16	15	8	7	0
0x50							
0x60							
0x70							
0x80							
Reset							

ISP_GAMMA_B_Y07 Configures the value of the 7th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y06 Configures the value of the 6th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y05 Configures the value of the 5th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y04 Configures the value of the 4th point on the Y axis of B channel gamma curve.
(R/W)

Register 36.34. ISP_GAMMA_BY3_REG (0x00A0)

<i>ISP_GAMMA_B_Y08</i>							
<i>ISP_GAMMA_B_Y09</i>							
<i>ISP_GAMMA_B_Y0A</i>							
<i>ISP_GAMMA_B_Y0B</i>							
31	24	23	16	15	8	7	0
0x90							
0xa0							
0xb0							
0xc0							
Reset							

ISP_GAMMA_B_Y0B Configures the value of the 11th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y0A Configures the value of the 10th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y09 Configures the value of the 9th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y08 Configures the value of the 8th point on the Y axis of B channel gamma curve.
(R/W)

Register 36.35. ISP_GAMMA_BY4_REG (0x00A4)

<i>ISP_GAMMA_B_Y0C</i>							
<i>ISP_GAMMA_B_Y0D</i>							
<i>ISP_GAMMA_B_Y0E</i>							
<i>ISP_GAMMA_B_Y0F</i>							
31	24	23	16	15	8	7	0
0xd0							
0xe0							
0xf0							
0xff							
Reset							

ISP_GAMMA_B_Y0F Configures the value of the 15th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y0E Configures the value of the 14th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y0D Configures the value of the 13th point on the Y axis of B channel gamma curve.
(R/W)

ISP_GAMMA_B_Y0C Configures the value of the 12th point on the Y axis of B channel gamma curve.
(R/W)

Register 36.36. ISP_GAMMA_RX1_REG (0x00A8)

(reserved)								ISP_GAMMA_R_X00		ISP_GAMMA_R_X01		ISP_GAMMA_R_X02		ISP_GAMMA_R_X03		ISP_GAMMA_R_X04		ISP_GAMMA_R_X05		ISP_GAMMA_R_X06		ISP_GAMMA_R_X07			
31								24	23	21	20	18	17	15	14	12	11	9	8	6	5	3	2	0	
0	0	0	0	0	0	0	0	0	4		4		4		4		4		4		4		4		Reset

ISP_GAMMA_R_X07 Configures the width of the 7th interval on the X axis of R channel gamma curve. Refer to section [36.5.2.8](#) for configuration. (R/W)

ISP_GAMMA_R_X06 Configures the width of the 6th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X05 Configures the width of the 5th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X04 Configures the width of the 4th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X03 Configures the width of the 3rd interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X02 Configures the width of the 2nd interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X01 Configures the width of the 1st interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X00 Configures the width of the 0th interval on the X axis of R channel gamma curve. (R/W)

Register 36.37. ISP_GAMMA_RX2_REG (0x00AC)

(reserved)								ISP_GAMMA_R_X08				ISP_GAMMA_R_X09				ISP_GAMMA_R_X0A				ISP_GAMMA_R_X0B				ISP_GAMMA_R_X0C				ISP_GAMMA_R_X0D				ISP_GAMMA_R_X0E				ISP_GAMMA_R_X0F																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
31	24	23	21	20	18	17	15	14	12	11	9	8	6	5	3	2	0	Reset																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

ISP_GAMMA_R_X0F Configures the width of the 15th interval on the X axis of R channel gamma curve. Refer to section [36.5.2.8](#) for configuration. (R/W)

ISP_GAMMA_R_X0E Configures the width of the 14th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X0D Configures the width of the 13th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X0C Configures the width of the 12th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X0B Configures the width of the 11th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X0A Configures the width of the 10th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X09 Configures the width of the 9th interval on the X axis of R channel gamma curve. (R/W)

ISP_GAMMA_R_X08 Configures the width of the 8th interval on the X axis of R channel gamma curve. (R/W)

Register 36.38. ISP_GAMMA_GX1_REG (0x00B0)

(reserved)								ISP_GAMMA_G_X00				ISP_GAMMA_G_X01				ISP_GAMMA_G_X02				ISP_GAMMA_G_X03				ISP_GAMMA_G_X04				ISP_GAMMA_G_X05				ISP_GAMMA_G_X06				ISP_GAMMA_G_X07													
31								24								23		21		20		18		17		15		14		12		11		9		8		6		5		3		2		0			
0								0								0		0		0		0		0		0		0		0		4		4		4		4		4		4		4		4		Reset	

Reset

ISP_GAMMA_G_X07 Configures the width of the 7th interval on the X axis of G channel gamma curve. Refer to section [36.5.2.8](#) for configuration. (R/W)

ISP_GAMMA_G_X06 Configures the width of the 6th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X05 Configures the width of the 5th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X04 Configures the width of the 4th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X03 Configures the width of the 3rd interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X02 Configures the width of the 2nd interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X01 Configures the width of the 1st interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X00 Configures the width of the 0th interval on the X axis of G channel gamma curve. (R/W)

Register 36.39. ISP_GAMMA_GX2_REG (0x00B4)

(reserved)								ISP_GAMMA_G_X08		ISP_GAMMA_G_X09		ISP_GAMMA_G_X0A		ISP_GAMMA_G_X0B		ISP_GAMMA_G_X0C		ISP_GAMMA_G_X0D		ISP_GAMMA_G_X0E		ISP_GAMMA_G_X0F	
31		24	23	21	20	18	17	15	14	12	11	9	8	6	5	3	2	0					
0	0	0	0	0	0	0	0	0	4		4		4		4		4		4		4		4
																							Reset

Reset

ISP_GAMMA_G_X0F Configures the width of the 15th interval on the X axis of G channel gamma curve. Refer to section [36.5.2.8](#) for configuration. (R/W)

ISP_GAMMA_G_X0E Configures the width of the 14th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X0D Configures the width of the 13th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X0C Configures the width of the 12th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X0B Configures the width of the 11th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X0A Configures the width of the 10th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X09 Configures the width of the 9th interval on the X axis of G channel gamma curve. (R/W)

ISP_GAMMA_G_X08 Configures the width of the 8th interval on the X axis of G channel gamma curve. (R/W)

Register 36.40. ISP_GAMMA_BX1_REG (0x00B8)

(reserved)								ISP_GAMMA_B_X00				ISP_GAMMA_B_X01				ISP_GAMMA_B_X02				ISP_GAMMA_B_X03				ISP_GAMMA_B_X04				ISP_GAMMA_B_X05				ISP_GAMMA_B_X06				ISP_GAMMA_B_X07			
31	24	23	21	20	18	17	15	14	12	11	9	8	6	5	3	2	0																						
0	0	0	0	0	0	0	0	0	4	4	4	4	4	4	4	4	4	Reset																					

ISP_GAMMA_B_X07 Configures the width of the 7th interval on the X axis of B channel gamma curve. Refer to section [36.5.2.8](#) for configuration. (R/W)

ISP_GAMMA_B_X06 Configures the width of the 6th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X05 Configures the width of the 5th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X04 Configures the width of the 4th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X03 Configures the width of the 3rd interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X02 Configures the width of the 2nd interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X01 Configures the width of the 1st interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X00 Configures the width of the 0th interval on the X axis of B channel gamma curve. (R/W)

Register 36.41. ISP_GAMMA_BX2_REG (0x00BC)

(reserved)								ISP_GAMMA_B_X08		ISP_GAMMA_B_X09		ISP_GAMMA_B_X0A		ISP_GAMMA_B_X0B		ISP_GAMMA_B_X0C		ISP_GAMMA_B_X0D		ISP_GAMMA_B_X0E		ISP_GAMMA_B_X0F			
31								24	23	21	20	18	17	15	14	12	11	9	8	6	5	3	2	0	
0	0	0	0	0	0	0	0	0	4		4		4		4		4		4		4		4		Reset

ISP_GAMMA_B_X0F Configures the width of the 15th interval on the X axis of B channel gamma curve. Refer to section 36.5.2.8 for configuration. (R/W)

ISP_GAMMA_B_X0E Configures the width of the 14th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X0D Configures the width of the 13th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X0C Configures the width of the 12th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X0B Configures the width of the 11th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X0A Configures the width of the 10th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X09 Configures the width of the 9th interval on the X axis of B channel gamma curve. (R/W)

ISP_GAMMA_B_X08 Configures the width of the 8th interval on the X axis of B channel gamma curve. (R/W)

Register 36.42. ISP_AE_CTRL_REG (0x00C0)

(reserved)																										ISP_AE_SELECT		ISP_AE_UPDATE	
31																											2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

ISP_AE_UPDATE Configures AE statistics sampling. With AE enabled, writing 1 triggers a statistics collection. (WT)

ISP_AE_SELECT Configures the AE sampling point.

0: Use data output from the demosaic module

1: Use data output from the gamma correction module

(R/W)

Register 36.43. ISP_AE_MONITOR_REG (0x00C4)

(reserved)										ISP_AE_MONITOR_PERIOD						ISP_AE_MONITOR_TH						ISP_AE_MONITOR_TL					
31											22	21				16	15				8	7				0	
0	0	0	0	0	0	0	0	0	0	0	0			0			0			0			Reset				

Reset

ISP_AE_MONITOR_TL Configures the lower threshold for AE luminance monitoring. (R/W)**ISP_AE_MONITOR_TH** Configures the upper threshold for AE luminance monitoring. (R/W)**ISP_AE_MONITOR_PERIOD** Configures the frame interval for AE monitoring. (R/W)**Register 36.44. ISP_AE_BX_REG (0x00C8)**

(reserved)											ISP_AE_X_START											ISP_AE_X_BSIZE										
31											22	21								11	10						0					
0	0	0	0	0	0	0	0	0	0	0	0										384										Reset	

Reset

ISP_AE_X_BSIZE Configures the horizontal size of each sub-window. (R/W)**ISP_AE_X_START** Configures the start coordinate of the AE statistical window in the horizontal direction. (R/W)**Register 36.45. ISP_AE_BY_REG (0x00CC)**

(reserved)												ISP_AE_Y_START												ISP_AE_Y_BSIZE											
31											22	21								11	10						0								
0	0	0	0	0	0	0	0	0	0	0	0										216										Reset				

Reset

ISP_AE_Y_BSIZE Configures the vertical dimension of each sub-window. (R/W)**ISP_AE_Y_START** Configures the start coordinate of the AE statistical window in the vertical direction. (R/W)

Register 36.46. ISP_AE_WINPIXNUM_REG (0x00D0)

(reserved)																ISP_AE_SUBWIN_PIXNUM															
31																17	16														0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	82944															Reset

ISP_AE_SUBWIN_PIXNUM Configures the number of pixels in each sub-window. (R/W)

Register 36.47. ISP_AE_WIN_RECIPROCAL_REG (0x00D4)

(reserved)																ISP_AE_SUBWIN_RECIP																																																																																																																																																															
31																20																19																0																																																																																																																															
0																0																0																0																0																0																0																0																0																0x000																Reset															

ISP_AE_SUBWIN_RECIP Configures the reciprocal of the number of pixels in each sub-window. It is a 20-bit fractional number, with [19:0] representing the fractional part. (R/W)

Register 36.48. ISP_SHARP_CTRL0_REG (0x00F4)

ISP_SHARP_AMOUNT_HIGH								ISP_SHARP_AMOUNT_LOW								ISP_SHARP_THRESHOLD_HIGH								ISP_SHARP_THRESHOLD_LOW									
31								24	23								16	15								8	7						0
0								0								0								0								Reset	

ISP_SHARP_THRESHOLD_LOW Configures the threshold for sharpening high-frequency details. Refer to section 36.5.2.10 for configuration. (R/W)

ISP_SHARP_THRESHOLD_HIGH Configures the threshold for sharpening high-frequency edges. Refer to section 36.5.2.10 for configuration. (R/W)

ISP_SHARP_AMOUNT_LOW Configures the gain for sharpening high-frequency details. Refer to section 36.5.2.10 for configuration. (R/W)

ISP_SHARP_AMOUNT_HIGH Configures the gain for sharpening high-frequency edges. Refer to section 36.5.2.10 for configuration. (R/W)

Register 36.49. ISP_SHARP_FILTER0_REG (0x00F8)

(reserved)																ISP_SHARP_FILTER_COE02				ISP_SHARP_FILTER_COE01				ISP_SHARP_FILTER_COE00			
31															15	14			10	9			5	4			0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		2		1							

Reset

ISP_SHARP_FILTER_COE00 Configures the value of coordinate 00 in the low-pass filter template of the sharpen module. (R/W)

ISP_SHARP_FILTER_COE01 Configures the value of coordinate 01 in the low-pass filter template of the sharpen module. (R/W)

ISP_SHARP_FILTER_COE02 Configures the value of coordinate 02 in the low-pass filter template of the sharpen module. (R/W)

Register 36.50. ISP_SHARP_FILTER1_REG (0x00FC)

(reserved)																ISP_SHARP_FILTER_COE12				ISP_SHARP_FILTER_COE11				ISP_SHARP_FILTER_COE10			
31															15	14			10	9			5	4			0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		4		2							

Reset

ISP_SHARP_FILTER_COE10 Configures the value of coordinate 10 in the low-pass filter template of the sharpen module. (R/W)

ISP_SHARP_FILTER_COE11 Configures the value of coordinate 11 in the low-pass filter template of the sharpen module. (R/W)

ISP_SHARP_FILTER_COE12 Configures the value of coordinate 12 in the low-pass filter template of the sharpen module. (R/W)

Register 36.51. ISP_SHARP_FILTER2_REG (0x0100)

(reserved)																ISP_SHARP_FILTER_COE22				ISP_SHARP_FILTER_COE21				ISP_SHARP_FILTER_COE20			
31															15	14			10	9			5	4			0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1			2			1					Reset

- ISP_SHARP_FILTER_COE20 Configures the value of coordinate 20 in the low-pass filter template of the sharpen module. (R/W)
- ISP_SHARP_FILTER_COE21 Configures the value of coordinate 21 in the low-pass filter template of the sharpen module. (R/W)
- ISP_SHARP_FILTER_COE22 Configures the value of coordinate 22 in the low-pass filter template of the sharpen module. (R/W)

Register 36.52. ISP_SHARP_MATRIX_CTRL_REG (0x0104)

(reserved)								ISP_SHARP_PADDING_MODE								ISP_SHARP_PADDING_DATA								ISP_SHARP_TAIL_PIXEN_PULSE_TH								ISP_SHARP_TAIL_PIXEN_PULSE_TL																																
31								25								24	23								16								15								8								7								0							
0 0 0 0 0 0 0 0								0								0								0x0								0x0								Reset																								

ISP_SHARP_TAIL_PIXEN_PULSE_TL Configures the data control cycle for the tail-row data rate during pixel-to-matrix conversion in the sharpen module.

Within [ISP_SHARP_TAIL_PIXEN_PULSE_TL](#) clock cycles, only the first [ISP_SHARP_TAIL_PIXEN_PULSE_TH](#) cycles are valid. Be cautious to configure this field, to prevent data processing from being deferred to the next frame.

This feature is enabled only when both [ISP_SHARP_TAIL_PIXEN_PULSE_TL](#) and [ISP_SHARP_TAIL_PIXEN_PULSE_TH](#) are non-zero, and [ISP_SHARP_TAIL_PIXEN_PULSE_TH](#) < [ISP_SHARP_TAIL_PIXEN_PULSE_TL](#). (R/W)

ISP_SHARP_TAIL_PIXEN_PULSE_TH Configures the valid data cycle of the tail-row data rate during pixel-to-matrix conversion in the sharpen module, which must be less than [ISP_HADR_NUM](#) - 1. (R/W)

ISP_SHARP_PADDING_DATA Configures the padding data used during image edge expansion in pixel-to-matrix conversion of the sharpen module. (R/W)

ISP_SHARP_PADDING_MODE Configures the method of image edge expansion during pixel-to-matrix conversion in the sharpen module.

0: Automatically padded with pixel values from the image edge

1: Padded with the data in [ISP_SHARP_PADDING_DATA](#)

(R/W)

Register 36.53. ISP_SHARP_CTRL1_REG (0x0108)

(reserved)																ISP_SHARP_GRADIENT_MAX															
31															8	7															0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0							0x0							Reset

ISP_SHARP_GRADIENT_MAX Represents the maximum pixel value of high-frequency components in the sharpen module, automatically refreshed every frame. (RO)

Register 36.54. ISP_DMA_CNTL_REG (0x010C)

ISP_DMA_INTERVAL										ISP_DMA_BURST_LEN										ISP_DMA_DATA_TYPE										ISP_DMA_UPDATE_REG																																																	
31										20										19										8										7										2										1										0									
0x1										0x80										0x2a										0										0										Reset																													

ISP_DMA_EN Configures whether to trigger VDMA transfer of one frame of data.

0: Not trigger

1: Trigger

(WT)

ISP_DMA_UPDATE_REG Configures whether to update the configuration for **ISP_DMA_DATA_TYPE**, **ISP_DMA_BURST_LEN**, and **ISP_DMA_INTERVAL**.

0: Not update

1: Update, will be automatically cleared to 0 after the update is complete

(R/W)

ISP_DMA_DATA_TYPE Configures the Image Interface data_type for the transferred data.

0x2A: RAW8

0x2B: RAW10

0x2C: RAW12

Others: Invalid

(R/W)

ISP_DMA_BURST_LEN Configures the burst length for one VDMA transfer, which needs to align with the VDMA configuration. (R/W)

ISP_DMA_INTERVAL Configures the interval of VDMA requests. 12'b1: 1 clock cycle, 12'b11: 2 clock cycles, etc. (R/W)

Register 36.55. ISP_DMA_RAW_DATA_REG (0x0110)

ISP_DMA_RAW_NUM_TOTAL_SET										(reserved)												ISP_DMA_RAW_NUM_TOTAL																						
31	30																					22	21																					0
0	0	0	0	0	0	0	0	0	0	0	0												Reset																					

Reset

ISP_DMA_RAW_NUM_TOTAL Configures the amount of 64-bit data in a frame of image. (R/W)

ISP_DMA_RAW_NUM_TOTAL_SET Configures whether to update the configuration for [ISP_DMA_RAW_NUM_TOTAL](#).

0: Not update

1: Update

(WT)

Register 36.56. ISP_CAM_CNTL_REG (0x0114)

(reserved)																																ISP_CAM_CLK_INV					ISP_CAM_RESET					ISP_CAM_UPDATE					ISP_CAM_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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Reset

ISP_CAM_EN Configures whether to enable the input via the DVP interface.

0: Disable

1: Enable

(R/W)

ISP_CAM_UPDATE_REG Configures whether to update the configuration for [ISP_CAM_CONF_REG](#).

0: Not update

1: Update

(R/W)

ISP_CAM_RESET Configures whether to reset the DVP input configuration.

0: Not reset

1: Reset

(R/W)

ISP_CAM_CLK_INV Configures whether to invert the DVP clock.

0: Not invert

1: Invert

(R/W)

Register 36.57. ISP_CAM_CONF_REG (0x0118)

(reserved)																ISP_CAM_DE_ONLY				ISP_CAM_VSYNC_FILTER_EN				ISP_CAM_VSYNC_FILTER_THRES				ISP_CAM_VSYNC_INV				ISP_CAM_HSYNC_INV				ISP_CAM_DE_INV				ISP_CAM_DATA_TYPE				ISP_CAM_2BYTE_MODE				ISP_CAM_DATA_ORDER			
31																16	15	14	13	11	10	9	8	7	2	1	0	Reset																							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0x2a	0	0																										

Reset

ISP_CAM_DATA_ORDER Configures the order of DVP input data.

0: cam_data_in

1: cam_data_in[7:0], cam_data_in[15:8]

(R/W)

ISP_CAM_2BYTE_MODE Configures whether to enable the 2-byte mode.

0: Disable

1: Enable

(R/W)

ISP_CAM_DATA_TYPE Configures the format of DVP input data.

0x2a: RAW8

0x2b: RAW10

0x2c: RAW12

(R/W)

ISP_CAM_DE_INV Configures whether to invert the DVP de signal.

0: Not invert

1: Invert

(R/W)

ISP_CAM_HSYNC_INV Configures whether to invert the DVP hsync signal.

0: Not invert

1: Invert

(R/W)

ISP_CAM_VSYNC_INV Configures whether to invert the DVP vsync signal.

0: Not invert

1: Invert

(R/W)

ISP_CAM_VSYNC_FILTER_THRES Configures vsync filtering. Vsyzns shorter than this length will be filtered out. (R/W)

Continued on the next page...

Register 36.57. ISP_CAM_CONF_REG (0x0118)

Continued from the previous page...

ISP_CAM_VSYNC_FILTER_EN Configures whether to enable the vsync filtering.

0: Disable

1: Enable

(R/W)

ISP_CAM_DE_ONLY Configures whether the DVP interface uses only DE data without HSYNC data.

0: HSYNC is present

1: HSYNC is not present

(R/W)

Register 36.58. ISP_AF_CTRL0_REG (0x011C)

(reserved)								ISP_AF_ENV_PERIOD								(reserved)				ISP_AF_ENV_THRESHOLD				(reserved)				ISP_AF_MANUAL_UPDATE				(reserved)				ISP_AF_AUTO_UPDATE																																							
31								24								23								16								15				12				11				8				7				5				4				3				1				0							
0								0								0								0								0				0				0				0				0				0				0				0				0				0				Reset			

Reset

ISP_AF_AUTO_UPDATE Configures whether to enable AF automatic statistics.

0: Disable automatic statistics, and manual trigger is required

1: Enable automatic statistics, and AF will perform statistics for every frame with the AF module enabled

(R/W)

ISP_AF_MANUAL_UPDATE Configures AF manual statistics. With AF automatic statistics disabled, writing 1 triggers a statistics collection. (WT)**ISP_AF_ENV_THRESHOLD** Configures the threshold for AF scene monitoring. If [ISP_AF_ENV_USER_THRESHOLD_SUM](#) or [ISP_AF_ENV_USER_THRESHOLD_LUM](#) is 0, an interrupt for scene monitoring will be triggered when the changes in sharpness or luminance exceed the product of the baseline value and this field for two consecutive times. 4 fractional bits. (R/W)**ISP_AF_ENV_PERIOD** Configures the statistical interval frames for AF scene monitoring. Setting this field to 0 disables the scene monitoring. (R/W)

ISP_AF_ENV_USER_THRESHOLD_SUM

ISP_AF_ENV_USER_THRESHOLD_LUM

```

(reserved)
ISP_AF_THRESHOLD

```

Register 36.62. ISP_AF_HSCALE_A_REG (0x0134)

(reserved)				ISP_AF_LPPOINT_A								(reserved)				ISP_AF_RPOINT_A								
31	28	27	16								15	12	11	0										
0	0	0	0	1								0	0	0	0	128								Reset

ISP_AF_RPOINT_A Configures the right boundary coordinate of AF statistical window A, recommended to be less than HNUM-1. (R/W)

ISP_AF_LPOINT_A Configures the left boundary coordinate of AF statistical window A, recommended to be greater than or equal to 2. (R/W)

Register 36.63. ISP_AF_VSCALE_A_REG (0x0138)

(reserved)				ISP_AF_IPPOINT_A								(reserved)				ISP_AF_BPPOINT_A												
31	28	27													16	15	12	11									0	
0	0	0	0	1												0	0	0	0	128								Reset

ISP_AF_BPOINT_A Configures the bottom boundary coordinate of AF statistical window A, recommended to be less than HNUM-1. (R/W)

ISP_AF_TPOINT_A Configures the top boundary coordinate of AF statistical window A, recommended to be greater than or equal to 2. (R/W)

Register 36.64. ISP_AF_HSCALE_B_REG (0x013C)

(reserved)				ISP_AF_LPOINT_B				(reserved)				ISP_AF_RPOINT_B			
31	28	27	16	15	12	11	0								
0	0	0	0	1	0	0	0	0	128	Reset					

ISP_AF_RPOINT_B Configures the right boundary coordinate of AF statistical window B, recommended to be less than HNUM-1. (R/W)

ISP_AF_LPOINT_B Configures the left boundary coordinate of AF statistical window B, recommended to be greater than or equal to 2. (R/W)

Register 36.65. ISP_AF_VSCALE_B_REG (0x0140)

(reserved)				ISP_AF_TPOINT_B								(reserved)				ISP_AF_BPOINT_B							
31	28	27					16	15		12	11												0
0	0	0	0				1			0	0	0	0										128

Reset

ISP_AF_BPOINT_B Configures the bottom boundary coordinate of AF statistical window B, recommended to be less than HNUM-1. (R/W)

ISP_AF_TPOINT_B Configures the top boundary coordinate of AF statistical window B, recommended to be greater than or equal to 2. (R/W)

Register 36.66. ISP_AF_HSCALE_C_REG (0x0144)

(reserved)				ISP_AF_LPOINT_C								(reserved)				ISP_AF_RPOINT_C							
31	28	27					16	15		12	11												0
0	0	0	0				1			0	0	0	0										128

Reset

ISP_AF_RPOINT_C Configures the right boundary coordinate of AF statistical window C, recommended to be less than HNUM-1. (R/W)

ISP_AF_LPOINT_C Configures the left boundary coordinate of AF statistical window C, recommended to be greater than or equal to 2. (R/W)

Register 36.67. ISP_AF_VSCALE_C_REG (0x0148)

(reserved)				ISP_AF_TPOINT_C								(reserved)				ISP_AF_BPOINT_C							
31	28	27					16	15		12	11												0
0	0	0	0				1			0	0	0	0										128

Reset

ISP_AF_BPOINT_C Configures the bottom boundary coordinate of AF statistical window C, recommended to be less than HNUM-1. (R/W)

ISP_AF_TPOINT_C Configures the top boundary coordinate of AF statistical window C, recommended to be greater than or equal to 2. (R/W)

Register 36.68. ISP_AWB_MODE_REG (0x0164)

(reserved)																										ISP_AWB_SAMPLE (reserved)				ISP_AWB_MODE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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Reset

ISP_AWB_MODE Configures whether to enable the AWB algorithm.

1: Enable

Others: Invalid

(R/W)

ISP_AWB_SAMPLE Configures the AWB sampling point.

0: Use data output before processed by CCM

1: Use data output after processed by CCM

(R/W)

Register 36.69. ISP_AWB_HSCALE_REG (0x0168)

(reserved)				ISP_AWB_LPOINT												(reserved)				ISP_AWB_RPOINT											
31	28			27								16	15	12			11					0									
0	0	0	0	0							0	0	0	0	1919				Reset												

Reset

ISP_AWB_RPOINT Configures the right boundary coordinate of the AWB statistical window. (R/W)**ISP_AWB_LPOINT** Configures the left boundary coordinate of the AWB statistical window. (R/W)**Register 36.70. ISP_AWB_VSCALE_REG (0x016C)**

(reserved)				ISP_AWB_TPOINT								(reserved)				ISP_AWB_BPOINT							
31	28			27								16	15	12			11					0	
0	0	0	0	0							0	0	0	0	1079							Reset	

Reset

ISP_AWB_BPOINT Configures the bottom boundary coordinate of the AWB statistical window.
(R/W)**ISP_AWB_TPOINT** Configures the top boundary coordinate of the AWB statistical window. (R/W)

Register 36.71. ISP_AWB_TH_LUM_REG (0x0170)

(reserved)						ISP_AWB_MAX_LUM										(reserved)						ISP_AWB_MIN_LUM												
31	26	25	16	15	10	9	0																											
0	0	0	0	0	0	0	765										0	0	0	0	0	0	0	0										Reset

Reset

ISP_AWB_MIN_LUM Configures the lower limit of R+G+B luminance for AWB white patch filtering. (R/W)

ISP_AWB_MAX_LUM Configures the upper limit of R+G+B luminance for AWB white patch filtering. (R/W)

Register 36.72. ISP_AWB_TH_RG_REG (0x0174)

(reserved)						ISP_AWB_MAX_RG										(reserved)						ISP_AWB_MIN_RG												
312625						1615										109						0												
000000						0x3ff										000000						0												Reset

Reset

ISP_AWB_MIN_RG Configures the lower limit of R/G ratio for AWB white point filtering. Bits [9:8] are the integer part, and bits [7:0] are the fractional part. (R/W)

ISP_AWB_MAX_RG Configures the upper limit of R/G ratio for AWB white point filtering. Bits [9:8] are the integer part, and bits [7:0] are the fractional part. (R/W)

Register 36.73. ISP_AWB_TH_BG_REG (0x0178)

(reserved)						ISP_AWB_MAX_BG										(reserved)						ISP_AWB_MIN_BG																																					
31						26						25						16						15						10						9						0																	
0						0						0						0						0						0						0x3ff						0						0						Reset					

Reset

ISP_AWB_MIN_BG Configures the lower limit of B/G ratio for AWB white point filtering. Bits [9:8] are the integer part, and bits [7:0] are the fractional part. (R/W)

ISP_AWB_MAX_BG Configures the upper limit of B/G ratio for AWB white point filtering. Bits [9:8] are the integer part, and bits [7:0] are the fractional part. (R/W)

Register 36.74. ISP_COLOR_CTRL_REG (0x018C)

ISP_COLOR_BRIGHTNESS							
ISP_COLOR_CONTRAST							
ISP_COLOR_HUE							
ISP_COLOR_SATURATION							
31	24	23	16	15	8	7	0
Reset							
0x0							
0x80							
0x0							
0x80							

ISP_COLOR_SATURATION Configures the saturation. Bit [7] is the integer part, and bits [6:0] are the fractional part. (R/W)

ISP_COLOR_HUE Configures the lower 8 bits of the hue value. The hue value is 9 bits in total, with the MSB provided by [ISP_COLOR_HUE_H](#). Valid range: 0–359. (R/W)

ISP_COLOR_CONTRAST Configures the saturation. Bit [7] is the integer part, and bits [6:0] are the fractional part. (R/W)

ISP_COLOR_BRIGHTNESS Configures the brightness in two's complement. (R/W)

Register 36.75. ISP_BLC_VALUE_REG (0x0190)

ISP_BLC_R0_VALUE							
ISP_BLC_R1_VALUE							
ISP_BLC_R2_VALUE							
ISP_BLC_R3_VALUE							
31	24	23	16	15	8	7	0
Reset							
0							
0							
0							
0							

ISP_BLC_R3_VALUE Configures the black level offset of channel 3 in the Bayer image. (R/W)

ISP_BLC_R2_VALUE Configures the black level offset of channel 2 in the Bayer image. (R/W)

ISP_BLC_R1_VALUE Configures the black level offset of channel 1 in the Bayer image. (R/W)

ISP_BLC_R0_VALUE Configures the black level offset of channel 0 in the Bayer image. (R/W)

Register 36.76. ISP_BLC_CTRL0_REG (0x0194)

(reserved)																												ISP_BLC_R0_STRETCH				ISP_BLC_R1_STRETCH				ISP_BLC_R2_STRETCH				ISP_BLC_R3_STRETCH																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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ISP_BLC_R3_STRETCH Configures whether to stretch the pixel range of channel 3.
0: Stretch
1: Not stretch
(R/W)

ISP_BLC_R2_STRETCH Configures whether to stretch the pixel range of channel 2.
0: Stretch
1: Not stretch
(R/W)

ISP_BLC_R1_STRETCH Configures whether to stretch the pixel range of channel 1.
0: Stretch
1: Not stretch
(R/W)

ISP_BLC_RO_STRETCH Configures whether to stretch the pixel range of channel 0.
0: Stretch
1: Not stretch
(R/W)

Register 36.77. ISP_HIST_MODE_REG (0x01A4)

[illegible]

ISP_HIST_MODE Configures the HIST sampling mode.

- 0: RAW_B
1: RAW_GB
2: RAW_GR
3: RAW_R
4: RGB
5: YUV_Y
6: YUV_U
7: YUV_V
(R/W)

Register 36.78. ISP_HIST_COEFF_REG (0x01A8)

(reserved)									ISP_HIST_COEFF_R								ISP_HIST_COEFF_G								ISP_HIST_COEFF_B										
31								24	23								16	15								8	7								0
0	0	0	0	0	0	0	0	0	85								85								85								Reset		

ISP_HIST_COEFF_B Configures the B weight for RGB to brightness conversion when HIST statistics mode set to RGB. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_COEFF_G Configures the G weight for RGB to brightness conversion when HIST statistics mode set to RGB. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_COEFF_R Configures the R weight for RGB to brightness conversion when HIST statistics mode set to RGB. Bits [7:0] are the fractional part. (R/W)

Register 36.79. ISP_HIST_OFFSETS_REG (0x01AC)

(reserved)				ISP_HIST_X_OFFSETS								(reserved)				ISP_HIST_Y_OFFSETS							
31	28	27					16	15	12	11					0								
0	0	0	0	0				0	0	0	0	0				Reset							

ISP_HIST_Y_OFFSETS Configures the start coordinate of the HIST statistics window in the vertical direction. (R/W)

ISP_HIST_X_OFFSETS Configures the start coordinate of the HIST statistics window in the horizontal direction. (R/W)

Register 36.80. ISP_HIST_SIZE_REG (0x01B0)

(reserved)								ISP_HIST_X_SIZE								(reserved)								ISP_HIST_Y_SIZE																																																																																																							
31								25								24								16								15								9								8								0																																																																							
0								0								0								0								0								0								0								18								0								0								0								0								0								0								32								Reset							

ISP_HIST_Y_SIZE Configures the vertical dimension of the HIST sub-window. (R/W)

ISP_HIST_X_SIZE Configures the horizontal dimension of the HIST sub-window. (R/W)

Register 36.81. ISP_HIST_SEGO_REG (0x01B4)

ISP_HIST_SEG_0_1				ISP_HIST_SEG_1_2				ISP_HIST_SEG_2_3				ISP_HIST_SEG_3_4				
31		24	23				16	15		8	7				0	
16				32				48				64				Reset

ISP_HIST_SEG_3_4 Configures the threshold between the HIST interval 3 and 4. (R/W)

ISP_HIST_SEG_2_3 Configures the threshold between the HIST interval 2 and 3. (R/W)

ISP_HIST_SEG_1_2 Configures the threshold between the HIST interval 1 and 2. (R/W)

ISP_HIST_SEG_0_1 Configures the threshold between the HIST interval 0 and 1. (R/W)

Register 36.82. ISP_HIST_SEG1_REG (0x01B8)

ISP_HIST_SEG_4_5							
ISP_HIST_SEG_5_6							
ISP_HIST_SEG_6_7							
ISP_HIST_SEG_7_8							
31	24	23	16	15	8	7	0
Reset							
80							
96							
112							
128							

ISP_HIST_SEG_7_8 Configures the threshold between the HIST interval 7 and 8. (R/W)

ISP_HIST_SEG_6_7 Configures the threshold between the HIST interval 6 and 7. (R/W)

ISP_HIST_SEG_5_6 Configures the threshold between the HIST interval 5 and 6. (R/W)

ISP_HIST_SEG_4_5 Configures the threshold between the HIST interval 4 and 5. (R/W)

Register 36.83. ISP_HIST_SEG2_REG (0x01BC)

ISP_HIST_SEG_8_9							
ISP_HIST_SEG_9_10							
ISP_HIST_SEG_10_11							
ISP_HIST_SEG_11_12							
31	24	23	16	15	8	7	0
Reset							
144							
160							
176							
192							

ISP_HIST_SEG_11_12 Configures the threshold between the HIST interval 11 and 12. (R/W)

ISP_HIST_SEG_10_11 Configures the threshold between the HIST interval 10 and 11. (R/W)

ISP_HIST_SEG_9_10 Configures the threshold between the HIST interval 9 and 10. (R/W)

ISP_HIST_SEG_8_9 Configures the threshold between the HIST interval 8 and 9. (R/W)

Register 36.84. ISP_HIST_SEG3_REG (0x01C0)

(reserved)							
ISP_HIST_SEG_12_13							
ISP_HIST_SEG_13_14							
ISP_HIST_SEG_14_15							
31	24	23	16	15	8	7	0
Reset							
0	0	0	0	0	0	0	0
208							
224							
240							

ISP_HIST_SEG_14_15 Configures the threshold between the HIST interval 14 and 15. (R/W)

ISP_HIST_SEG_13_14 Configures the threshold between the HIST interval 13 and 14. (R/W)

ISP_HIST_SEG_12_13 Configures the threshold between the HIST interval 12 and 13. (R/W)

Register 36.85. ISP_HIST_WEIGHT0_REG (0x01C4)

ISP_HIST_WEIGHT_00							
ISP_HIST_WEIGHT_01							
ISP_HIST_WEIGHT_02							
ISP_HIST_WEIGHT_03							
31	24	23	16	15	8	7	0
1							
Reset							

ISP_HIST_WEIGHT_03 Configures the weight of HIST sub-window 03. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_02 Configures the weight of HIST sub-window 02. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_01 Configures the weight of HIST sub-window 01. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_00 Configures the weight of HIST sub-window 00. Bits [7:0] are the fractional part. (R/W)

Register 36.86. ISP_HIST_WEIGHT1_REG (0x01C8)

ISP_HIST_WEIGHT_04							
ISP_HIST_WEIGHT_10							
ISP_HIST_WEIGHT_11							
ISP_HIST_WEIGHT_12							
31	24	23	16	15	8	7	0
1							
Reset							

ISP_HIST_WEIGHT_12 Configures the weight of HIST sub-window 12. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_11 Configures the weight of HIST sub-window 11. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_10 Configures the weight of HIST sub-window 10. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_04 Configures the weight of HIST sub-window 04. Bits [7:0] are the fractional part. (R/W)

Register 36.87. ISP_HIST_WEIGHT2_REG (0x01CC)

ISP_HIST_WEIGHT_13							
ISP_HIST_WEIGHT_14							
ISP_HIST_WEIGHT_20							
ISP_HIST_WEIGHT_21							
31	24	23	16	15	8	7	0
1		1		1		1	
Reset							

ISP_HIST_WEIGHT_21 Configures the weight of HIST sub-window 21. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_20 Configures the weight of HIST sub-window 20. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_14 Configures the weight of HIST sub-window 14. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_13 Configures the weight of HIST sub-window 13. Bits [7:0] are the fractional part. (R/W)

Register 36.88. ISP_HIST_WEIGHT3_REG (0x01D0)

ISP_HIST_WEIGHT_22							
ISP_HIST_WEIGHT_23							
ISP_HIST_WEIGHT_24							
ISP_HIST_WEIGHT_30							
31	24	23	16	15	8	7	0
232		1		1		1	
Reset							

ISP_HIST_WEIGHT_30 Configures the weight of HIST sub-window 30. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_24 Configures the weight of HIST sub-window 24. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_23 Configures the weight of HIST sub-window 23. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_22 Configures the weight of HIST sub-window 22. Bits [7:0] are the fractional part. (R/W)

Register 36.89. ISP_HIST_WEIGHT4_REG (0x01D4)

ISP_HIST_WEIGHT_31							
ISP_HIST_WEIGHT_32							
ISP_HIST_WEIGHT_33							
ISP_HIST_WEIGHT_34							
31	24	23	16	15	8	7	0
1							
Reset							

ISP_HIST_WEIGHT_34 Configures the weight of HIST sub-window 34. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_33 Configures the weight of HIST sub-window 33. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_32 Configures the weight of HIST sub-window 32. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_31 Configures the weight of HIST sub-window 31. Bits [7:0] are the fractional part. (R/W)

Register 36.90. ISP_HIST_WEIGHT5_REG (0x01D8)

ISP_HIST_WEIGHT_40							
ISP_HIST_WEIGHT_41							
ISP_HIST_WEIGHT_42							
ISP_HIST_WEIGHT_43							
31	24	23	16	15	8	7	0
1							
Reset							

ISP_HIST_WEIGHT_43 Configures the weight of HIST sub-window 43. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_42 Configures the weight of HIST sub-window 42. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_41 Configures the weight of HIST sub-window 41. Bits [7:0] are the fractional part. (R/W)

ISP_HIST_WEIGHT_40 Configures the weight of HIST sub-window 40. Bits [7:0] are the fractional part. (R/W)

Register 36.91. ISP_HIST_WEIGHT6_REG (0x01DC)

(reserved)																								ISP_HIST_WEIGHT_44									
31																								8	7	0							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	Reset			

ISP_HIST_WEIGHT_44 Configures the weight of HIST sub-window 44. Bits [7:0] are the fractional part. (R/W)

Register 36.92. ISP_YUV_FORMAT_REG (0x0234)

(reserved)																															ISP_YUV_RANGE		ISP_YUV_MODE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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ISP_YUV_MODE Configures the YUV output mode.

0: ITU-R BT.601

1: ITU-R BT.709

(R/W)

ISP_YUV_RANGE Configures the YUV output range.

0: Full range (YUV2RGB)

1: Limit range (YUV_Limit)

(R/W)

Register 36.93. ISP_CROP_CTRL_REG (0x0244)

(reserved)																																ISP_CROP_SF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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ISP_CROP_SFT_RST Configures whether to reset the CROP error status.

0: Not reset

1: Reset

(WT)

Register 36.94. ISP_CROP_Y_CAPTURE_REG (0x0248)

(reserved)								ISP_CROP_Y_END								ISP_CROP_Y_START																																																															
31								24								23								12								11								0																																							
0								0								0								0								0								0								0								0x0								0x0								Reset							

Reset

ISP_CROP_Y_START Configures the start coordinate in the Y direction. (R/W)**ISP_CROP_Y_END** Configures the end coordinate in the Y direction. (R/W)**Register 36.95. ISP_CROP_X_CAPTURE_REG (0x024C)**

(reserved)								ISP_CROP_X_END								ISP_CROP_X_START																																																															
31								24								23								12								11								0																																							
0								0								0								0								0								0								0								0x0								0x0								Reset							

Reset

ISP_CROP_X_START Configures the start coordinate in the X direction. (R/W)**ISP_CROP_X_END** Configures the end coordinate in the X direction. (R/W)**Register 36.96. ISP_CROP_ERR_ST_REG (0x0250)**

(reserved)																ISP_CROP_X_START_ODD		ISP_CROP_Y_START_ODD		ISP_CROP_X_END_ODD		ISP_CROP_Y_END_ODD		ISP_CROP_X_END_EVEN		ISP_CROP_Y_MISMATCH		ISP_CROP_Y_MISMATCH	
31																	6	5	4	3	2	1	0						
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

ISP_CROP_Y_MISMATCH Represents an error where the configured CROP Y coordinate exceeds the image boundary. (RO)**ISP_CROP_X_MISMATCH** Represents an error where the configured CROP X coordinate exceeds the image boundary. (RO)**ISP_CROP_Y_END_EVEN** Represents an error where the Y-direction end coordinate is even. (RO)**ISP_CROP_X_END_EVEN** Represents an error where the X-direction end coordinate is even. (RO)**ISP_CROP_Y_START_ODD** Represents an error where the Y-direction start coordinate is odd. (RO)**ISP_CROP_X_START_ODD** Represents an error where the X-direction start coordinate is odd. (RO)

Register 36.97. ISP_WBG_COEF_R_REG (0x0254)

(reserved)																ISP_WBG_R																																															
31																12																11																0															
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ISP_WBG_R Configures the WBG R-channel gain: 4-bit integer and 8-bit fractional. (R/W)

Register 36.98. ISP_WBG_COEF_G_REG (0x0258)

(reserved)																				ISP_WBG_G																											
31												12								11												0															
0												0								0												256								Reset							

ISP_WBG_G Configures the WBG G-channel gain: 4-bit integer and 8-bit fractional. (R/W)

Register 36.99. ISP_WBG_COEF_B_REG (0x025C)

(reserved)																ISP_WBG_B																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					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ISP_WBG_B Configures the WBG B-channel gain: 4-bit integer and 8-bit fractional. (R/W)

Register 36.100. ISP_COLOR_HUE_CTRL_REG (0x0260)

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ISP_COLOR_HUE_H Configures the most significant bit of HUE. (R/W)

Register 36.101. ISP_AWB_BX_REG (0x0264)

(reserved)																ISP_AWB_X_START																ISP_AWB_X_BSIZE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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ISP_AWB_X_BSIZE Configures the AWB sub-window size in the X direction; minimum is 4. (R/W)

ISP_AWB_X_START Configures the AWB sub-window start coordinate in the X direction. (R/W)

Register 36.102. ISP_AWB_BY_REG (0x0268)

(reserved)								ISP_AWB_Y_START				ISP_AWB_Y_BSIZE			
312423								1211				0			
00000000								0				0Reset			

ISP_AWB_Y_BSIZE Configures the AWB sub-window size in the Y direction. (R/W)

ISP_AWB_Y_START Configures the AWB sub-window start coordinate in the Y direction. (R/W)

Register 36.103. ISP_STATE_REG (0x026C)

(reserved)																												ISP_HEADER_BUSY		ISP_TAIL_BUSY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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ISP_TAIL_BUSY Represents whether isp_tail is busy.

0: Idle

1: Busy

(RO)

ISP_HEADER_BUSY Represents whether isp_header is busy.

0: Idle

1: Busy

(RO)

Register 36.104. ISP_SHADOW_REG_CTRL_REG (0x0270)

ISP_SHADOW_UPDATE_SEL																								(reserved)								ISP_COLOR_UPDATE ISP_SHARP_UPDATE (reserved) ISP_CCM_UPDATE ISP_WBG_UPDATE ISP_BF_UPDATE ISP_DPC_UPDATE ISP_BLC_UPDATE							
31	30	29																	8	7	6	5	4	3	2	1	0	Reset											
0x1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0															

ISP_BLC_UPDATE Write 1 to update the BLC registers. (R/W)

ISP_DPC_UPDATE Write 1 to update the DPC registers. (R/W)

ISP_BF_UPDATE Write 1 to update the BF registers. (R/W)

ISP_WBG_UPDATE Write 1 to update the WBG registers. (R/W)

ISP_CCM_UPDATE Write 1 to update the CCM registers. (R/W)

ISP_SHARP_UPDATE Write 1 to update the SHARP registers. (R/W)

ISP_COLOR_UPDATE Write 1 to update the COLOR registers. (R/W)

ISP_SHADOW_UPDATE_SEL Configures the shadow register update mode.

0: Disable shadow registers

1: Automatically update at every vsync

2: After writing 1 to the corresponding update register, automatically update at the next vsync
(R/W)

Register 36.105. ISP_DPC_DEADPIX_CNT_REG (0x0044)

(reserved)																						ISP_DPC_DEADPIX_CNT																			
31																						9										0									
0 0																						0x0										Reset									

ISP_DPC_DEADPIX_CNT Represents the number of dead pixels counted during static calibration.
(RO)

Register 36.106. ISP_LUT_RDATA_REG (0x0050)

<i>ISP_LUT_RDATA</i>	
31	0
0x000000	
Reset	

ISP_LUT_RDATA Data read from the LUT. Read from this field after configuring [ISP_LUT_CMD](#). (RO)

Register 36.107. ISP_AE_BLOCK_MEAN_O_REG (0x00D8)

ISP_AE_B00_MEAN							
ISP_AE_B01_MEAN							
ISP_AE_B02_MEAN							
ISP_AE_B03_MEAN							
31	24	23	16	15	8	7	0
0		0		0		0	
Reset							

ISP_AE_B03_MEAN Represents the statistics from AE sub-window 03. (RO)

ISP_AE_B02_MEAN Represents the statistics from AE sub-window 02. (RO)

ISP_AE_B01_MEAN Represents the statistics from AE sub-window 01. (RO)

ISP_AE_B00_MEAN Represents the statistics from AE sub-window 00. (RO)

Register 36.108. ISP_AE_BLOCK_MEAN_1_REG (0x00DC)

ISP_AE_B04_MEAN							
ISP_AE_B10_MEAN							
ISP_AE_B11_MEAN							
ISP_AE_B12_MEAN							
31	24	23	16	15	8	7	0
0		0		0		0	
Reset							

ISP_AE_B12_MEAN Represents the statistics from AE sub-window 12. (RO)

ISP_AE_B11_MEAN Represents the statistics from AE sub-window 11. (RO)

ISP_AE_B10_MEAN Represents the statistics from AE sub-window 10. (RO)

ISP_AE_B04_MEAN Represents the statistics from AE sub-window 04. (RO)

Register 36.109. ISP_AE_BLOCK_MEAN_2_REG (0x00E0)

ISP_AE_B13_MEAN							
ISP_AE_B14_MEAN							
ISP_AE_B20_MEAN							
ISP_AE_B21_MEAN							
31	24	23	16	15	8	7	0
0							

Reset

ISP_AE_B21_MEAN Represents the statistics from AE sub-window 21. (RO)

ISP_AE_B20_MEAN Represents the statistics from AE sub-window 20. (RO)

ISP_AE_B14_MEAN Represents the statistics from AE sub-window 14. (RO)

ISP_AE_B13_MEAN Represents the statistics from AE sub-window 13. (RO)

Register 36.110. ISP_AE_BLOCK_MEAN_3_REG (0x00E4)

ISP_AE_B22_MEAN							
ISP_AE_B23_MEAN							
ISP_AE_B24_MEAN							
ISP_AE_B30_MEAN							
31	24	23	16	15	8	7	0
0							

Reset

ISP_AE_B30_MEAN Represents the statistics from AE sub-window 30. (RO)

ISP_AE_B24_MEAN Represents the statistics from AE sub-window 24. (RO)

ISP_AE_B23_MEAN Represents the statistics from AE sub-window 23. (RO)

ISP_AE_B22_MEAN Represents the statistics from AE sub-window 22. (RO)

Register 36.111. ISP_AE_BLOCK_MEAN_4_REG (0x00E8)

ISP_AE_B31_MEAN							
ISP_AE_B32_MEAN							
ISP_AE_B33_MEAN							
ISP_AE_B34_MEAN							
31	24	23	16	15	8	7	0
0							

Reset

ISP_AE_B34_MEAN Represents the statistics from AE sub-window 34. (RO)

ISP_AE_B33_MEAN Represents the statistics from AE sub-window 33. (RO)

ISP_AE_B32_MEAN Represents the statistics from AE sub-window 32. (RO)

ISP_AE_B31_MEAN Represents the statistics from AE sub-window 31. (RO)

Register 36.112. ISP_AE_BLOCK_MEAN_5_REG (0x00EC)

ISP_AE_B40_MEAN							
ISP_AE_B41_MEAN							
ISP_AE_B42_MEAN							
ISP_AE_B43_MEAN							
31	24	23	16	15	8	7	0
0							
Reset							

ISP_AE_B43_MEAN Represents the statistics from AE sub-window 43. (RO)

ISP_AE_B42_MEAN Represents the statistics from AE sub-window 42. (RO)

ISP_AE_B41_MEAN Represents the statistics from AE sub-window 41. (RO)

ISP_AE_B40_MEAN Represents the statistics from AE sub-window 40. (RO)

Register 36.113. ISP_AE_BLOCK_MEAN_6_REG (0x00F0)

ISP_AE_B44_MEAN																							
(reserved)																							
31	24	23																					0
0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Reset																							

ISP_AE_B44_MEAN Represents the statistics from AE sub-window 44. (RO)

Register 36.114. ISP_AF_SUM_A_REG (0x014C)

(reserved)		ISP_AF_SUMA																										
31	30	29																									0	
0	0	0x000000																										Reset

ISP_AF_SUMA Represents the sharpness statistics from AF window A. (RO)

Register 36.115. ISP_AF_SUM_B_REG (0x0150)

(reserved)			ISP_AF_SUMB																									
31	30	29	0																									
0	0	0x000000																										Reset

ISP_AF_SUMB Represents the sharpness statistics from AF window B. (RO)

Register 36.116. ISP_AF_SUM_C_REG (0x0154)

(reserved)			ISP_AF_SUMC																									
31	30	29	0																									
0	0	0x000000																										Reset

ISP_AF_SUMC Represents the sharpness statistics from AF window C. (RO)

Register 36.117. ISP_AF_LUM_A_REG (0x0158)

(reserved)				ISP_AF_LUMA																																																
31				28																								27																								0
0	0	0	0	0																																																

Reset

ISP_AF_LUMA Represents the luminance statistics from AF window A. (RO)

Register 36.118. ISP_AF_LUM_B_REG (0x015C)

(reserved)				ISP_AF_LUMB																									
31				28	27																								0
0	0	0	0	0																									Reset

ISP_AF_LUMB Represents the luminance statistics from AF window B. (RO)

Register 36.119. ISP_AF_LUM_C_REG (0x0160)

(reserved)				ISP_AF_LUM_C																								
31	28			27																								0
0	0	0	0		0																							Reset

ISP_AF_LUM_C Represents the luminance statistics from AF window C. (RO)

Register 36.120. ISP_AWBO_WHITE_CNT_REG (0x017C)

(reserved)								ISP_AWBO_WHITE_CNT																															
31								24								23																							0
0 0 0 0 0 0 0 0								0																							Reset								

ISP_AWBO_WHITE_CNT Represents the number of white patches within the AWB statistical window. (RO)

Register 36.121. ISP_AWBO_ACC_R_REG (0x0180)

ISP_AWBO_ACC_R																															

ISP_AWBO_ACC_R Represents the accumulated value of the R component for white patches within the AWB statistical window. (RO)

Register 36.122. ISP_AWBO_ACC_G_REG (0x0184)

ISP_AWBO_ACC_G																															
0																															
0																															
Reset																															

ISP_AWBO_ACC_G Represents the accumulated value of the G component for white patches within the AWB statistical window. (RO)

Register 36.123. ISP_AWBO_ACC_B_REG (0x0188)

ISP_AWBO_ACC_B																															
31																														0	
0																															
Reset																															

ISP_AWBO_ACC_B Represents the accumulated value of the B component for white patches within the AWB statistical window. (RO)

Register 36.124. ISP_HIST_BINO_REG (0x01E0)

(reserved)																	ISP_HIST_BIN_O														
311716																	0														
0000000000000000																	0														Reset

ISP_HIST_BIN_0 Represents the number of pixels where the luminance falls in the range of interval 0 in the HIST statistical window. (RO)

Register 36.125. ISP_HIST_BIN1_REG (0x01E4)

(reserved)																	ISP_HIST_BIN_1														
3117																	160														
0000000000000000																	0														Reset

ISP_HIST_BIN_1 Represents the number of pixels where the luminance falls in the range of interval 1 in the HIST statistical window. (RO)

Register 36.126. ISP_HIST_BIN2_REG (0x01E8)

(reserved)																	ISP_HIST_BIN_2															
31																	17	16														0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																	0														Reset	

ISP_HIST_BIN_2 Represents the number of pixels where the luminance falls in the range of interval 2 in the HIST statistical window. (RO)

Register 36.127. ISP_HIST_BIN3_REG (0x01EC)

(reserved)																ISP_HIST_BIN_3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																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Reset

ISP_HIST_BIN_3 Represents the number of pixels where the luminance falls in the range of interval 3 in the HIST statistical window. (RO)

Register 36.128. ISP_HIST_BIN4_REG (0x01F0)

(reserved)																ISP_HIST_BIN_4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																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Reset

ISP_HIST_BIN_5 Represents the number of pixels where the luminance falls in the range of interval 5 in the HIST statistical window. (RO)

Register 36.130. ISP_HIST_BIN6_REG (0x01F8)

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ISP_HIST_BIN_6 Represents the number of pixels where the luminance falls in the range of interval 6 in the HIST statistical window. (RO)

Register 36.131. ISP_HIST_BIN7_REG (0x01FC)

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ISP_HIST_BIN_7 Represents the number of pixels where the luminance falls in the range of interval 7 in the HIST statistical window. (RO)

Register 36.132. ISP_HIST_BIN8_REG (0x0200)

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ISP_HIST_BIN_8 Represents the number of pixels where the luminance falls in the range of interval 8 in the HIST statistical window. (RO)

Register 36.133. ISP_HIST_BIN9_REG (0x0204)

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ISP_HIST_BIN_10 Represents the number of pixels where the luminance falls in the range of interval 10 in the HIST statistical window. (RO)

Register 36.135. ISP_HIST_BIN11_REG (0x020C)

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ISP_HIST_BIN_11 Represents the number of pixels where the luminance falls in the range of interval 11 in the HIST statistical window. (RO)

Register 36.136. ISP_HIST_BIN12_REG (0x0210)

(reserved)																ISP_HIST_BIN_12																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															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ISP_HIST_BIN_12 Represents the number of pixels where the luminance falls in the range of interval 12 in the HIST statistical window. (RO)

Register 36.137. ISP_HIST_BIN13_REG (0x0214)

(reserved)															ISP_HIST_BIN_13																	
31															17	16																0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															0																Reset	

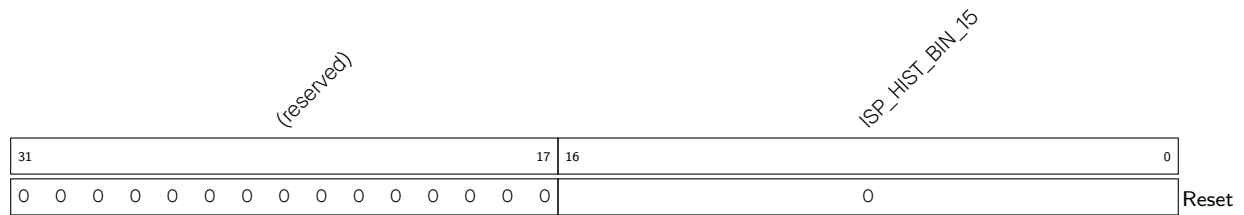
ISP_HIST_BIN_13 Represents the number of pixels where the luminance falls in the range of interval 13 in the HIST statistical window. (RO)

Register 36.138. ISP_HIST_BIN14_REG (0x0218)

(reserved)															ISP_HIST_BIN_14																	
31															17	16																0
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ISP_HIST_BIN_14 Represents the number of pixels where the luminance falls in the range of interval 14 in the HIST statistical window. (RO)

Register 36.139. ISP_HIST_BIN15_REG (0x021C)



ISP_HIST_BIN_15 Represents the number of pixels where the luminance falls in the range of interval 15 in the HIST statistical window. (RO)

Register 36.140. ISP_INT_RAW_REG (0x0064)

ISP_CROP_ERR_INT_RAW																																ISP_WBG_FRAME_INT_RAW																																ISP_CROP_FRAME_INT_RAW																																ISP_HEADER_IDI_FRAME_INT_RAW																																ISP_TAIL_IDI_FRAME_INT_RAW																																ISP_YUV2RGB_FRAME_INT_RAW																																ISP_COLOR_FRAME_INT_RAW																																ISP_SHARP_FRAME_INT_RAW																																ISP_RGB2YUV_FRAME_INT_RAW																																ISP_GAMMA_FRAME_INT_RAW																																(reserved)																																ISP_CCM_FRAME_INT_RAW																																ISP_DEMOSAIC_FRAME_INT_RAW																																ISP_BF_FRAME_INT_RAW																																ISP_DPC_FRAME_INT_RAW																																ISP_LSC_FRAME_INT_RAW																																ISP_BLC_FRAME_INT_RAW																																ISP_FRAME_INT_RAW																																ISP_HIST_FDONE_INT_RAW																																ISP_AWB_FDONE_INT_RAW																																ISP_AF_FDONE_INT_RAW																																ISP_AF_ENV_INT_RAW																																ISP_AE_FDONE_INT_RAW																																ISP_AE_MONITOR_INT_RAW																																ISP_GAMMA_XCOORD_ERR_INT_RAW																																ISP_DPC_CHECK_DONE_INT_RAW																																ISP_MIPI_HNUM_UNMATCH_INT_RAW																																ISP_HVNUM_SETTING_ERR_INT_RAW																																ISP_BUF_FULL_INT_RAW																																ISP_ASYNC_FIFO_OVF_INT_RAW																																ISP_DATA_TYPE_ERR_INT_RAW																																																																																											
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ISP_DATA_TYPE_ERR_INT_RAW The raw interrupt status of ISP_DATA_TYPE_ERR_INT. (R/SS/WTC)

ISP_ASYNC_FIFO_OVF_INT_RAW The raw interrupt status of ISP_ASYNC_FIFO_OVF_INT. (R/SS/WTC)

ISP_BUF_FULL_INT_RAW The raw interrupt status of ISP_BUF_FULL_INT. (R/SS/WTC)

ISP_HVNUM_SETTING_ERR_INT_RAW The raw interrupt status of ISP_HVNUM_SETTING_ERR_INT. (R/SS/WTC)

ISP_DATA_TYPE_SETTING_ERR_INT_RAW The raw interrupt status of ISP_DATA_TYPE_SETTING_ERR_INT. (R/SS/WTC)

ISP_MIPI_HNUM_UNMATCH_INT_RAW The raw interrupt status of ISP_MIPI_HNUM_UNMATCH_INT. (R/SS/WTC)

ISP_DPC_CHECK_DONE_INT_RAW The raw interrupt status of ISP_DPC_CHECK_DONE_INT. (R/SS/WTC)

ISP_GAMMA_XCOORD_ERR_INT_RAW The raw interrupt status of ISP_GAMMA_XCOORD_ERR_INT. (R/SS/WTC)

ISP_AE_MONITOR_INT_RAW The raw interrupt status of ISP_AE_MONITOR_INT. (R/SS/WTC)

ISP_AE_FRAME_DONE_INT_RAW The raw interrupt status of ISP_AE_FRAME_DONE_INT. (R/SS/WTC)

ISP_AF_FDONE_INT_RAW The raw interrupt status of ISP_AF_FDONE_INT. (R/SS/WTC)

ISP_AF_ENV_INT_RAW The raw interrupt status of ISP_AF_ENV_INT. (R/SS/WTC)

ISP_AWB_FDONE_INT_RAW The raw interrupt status of ISP_AWB_FDONE_INT. (R/SS/WTC)

ISP_HIST_FDONE_INT_RAW The raw interrupt status of ISP_HIST_FDONE_INT. (R/SS/WTC)

ISP_FRAME_INT_RAW The raw interrupt status of ISP_FRAME_INT. (R/SS/WTC)

ISP_BLC_FRAME_INT_RAW The raw interrupt status of ISP_BLC_FRAME_INT. (R/SS/WTC)

ISP_LSC_FRAME_INT_RAW The raw interrupt status of ISP_LSC_FRAME_INT. (R/SS/WTC)

Continued on the next page...

Register 36.140. ISP_INT_RAW_REG (0x0064)

Continued from the previous page...

ISP_DPC_FRAME_INT_RAW The raw interrupt status of ISP_DPC_FRAME_INT. (R/SS/WTC)

ISP_BF_FRAME_INT_RAW The raw interrupt status of ISP_BF_FRAME_INT. (R/SS/WTC)

ISP_DEMOSAIC_FRAME_INT_RAW The raw interrupt status of ISP_DEMOSAIC_FRAME_INT.
(R/SS/WTC)

ISP_CCM_FRAME_INT_RAW The raw interrupt status of ISP_CCM_FRAME_INT. (R/SS/WTC)

ISP_GAMMA_FRAME_INT_RAW The raw interrupt status of ISP_GAMMA_FRAME_INT. (R/SS/WTC)

ISP_RGB2YUV_FRAME_INT_RAW The raw interrupt status of ISP_RGB2YUV_FRAME_INT.
(R/SS/WTC)

ISP_SHARP_FRAME_INT_RAW The raw interrupt status of ISP_SHARP_FRAME_INT. (R/SS/WTC)

ISP_COLOR_FRAME_INT_RAW The raw interrupt status of ISP_COLOR_FRAME_INT. (R/SS/WTC)

ISP_YUV2RGB_FRAME_INT_RAW The raw interrupt status of ISP_YUV2RGB_FRAME_INT.
(R/SS/WTC)

ISP_TAIL_IDI_FRAME_INT_RAW The raw interrupt status of ISP_TAIL_IDI_FRAME_INT. (R/SS/WTC)

ISP_HEADER_IDI_FRAME_INT_RAW The raw interrupt status of ISP_HEADER_IDI_FRAME_INT.
(R/SS/WTC)

ISP_CROP_FRAME_INT_RAW The raw interrupt status of ISP_CROP_FRAME_INT. (R/SS/WTC)

ISP_WBG_FRAME_INT_RAW The raw interrupt status of ISP_WBG_FRAME_INT. (R/SS/WTC)

ISP_CROP_ERR_INT_RAW The raw interrupt status of ISP_CROP_ERR_INT. (R/SS/WTC)

Register 36.141. ISP_INT_ST_REG (0x0068)

ISP_CROP_ERR_INT_ST																																ISP_WBG_FRAME_INT_ST																																ISP_CROP_FRAME_INT_ST																																ISP_HEADER_IDI_FRAME_INT_ST																																ISP_TAIL_IDI_FRAME_INT_ST																																ISP_YUV2RGB_FRAME_INT_ST																																ISP_COLOR_FRAME_INT_ST																																ISP_SHARP_FRAME_INT_ST																																ISP_RGB2YUV_FRAME_INT_ST																																ISP_GAMMA_FRAME_INT_ST																																(reserved)																																ISP_CCM_FRAME_INT_ST																																ISP_DEMOSAIC_FRAME_INT_ST																																ISP_BF_FRAME_INT_ST																																ISP_DPC_FRAME_INT_ST																																ISP_LSC_FRAME_INT_ST																																ISP_BLC_FRAME_INT_ST																																ISP_FRAME_INT_ST																																ISP_HIST_FDONE_INT_ST																																ISP_AWB_FDONE_INT_ST																																ISP_AF_ENV_INT_ST																																ISP_AF_FDONE_INT_ST																																ISP_AE_FRAME_INT_ST																																ISP_AE_MONITOR_INT_ST																																ISP_GAMMA_XCOORD_ERR_INT_ST																																ISP_DPC_CHECK_DONE_INT_ST																																ISP_MIPI_HNUM_UNMATCH_INT_ST																																ISP_HVNUM_SETTING_ERR_INT_ST																																ISP_BUF_FULL_INT_ST																																ISP_ASYNC_FIFO_OVF_INT_ST																																ISP_DATA_TYPE_ERR_INT_ST																																																																																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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Reset

ISP_DATA_TYPE_ERR_INT_ST The masked interrupt status of ISP_DATA_TYPE_ERR_INT. (RO)**ISP_ASYNC_FIFO_OVF_INT_ST** The masked interrupt status of ISP_ASYNC_FIFO_OVF_INT. (RO)**ISP_BUF_FULL_INT_ST** The masked interrupt status of ISP_BUF_FULL_INT. (RO)**ISP_HVNUM_SETTING_ERR_INT_ST** The masked interrupt status of ISP_HVNUM_SETTING_ERR_INT. (RO)**ISP_DATA_TYPE_SETTING_ERR_INT_ST** The masked interrupt status of ISP_DATA_TYPE_SETTING_ERR_INT. (RO)**ISP_MIPI_HNUM_UNMATCH_INT_ST** The masked interrupt status of ISP_MIPI_HNUM_UNMATCH_INT. (RO)**ISP_DPC_CHECK_DONE_INT_ST** The masked interrupt status of ISP_DPC_CHECK_DONE_INT. (RO)**ISP_GAMMA_XCOORD_ERR_INT_ST** The masked interrupt status of ISP_GAMMA_XCOORD_ERR_INT. (RO)**ISP_AE_MONITOR_INT_ST** The masked interrupt status of ISP_AE_MONITOR_INT. (RO)**ISP_AE_FRAME_DONE_INT_ST** The masked interrupt status of ISP_AE_FRAME_DONE_INT. (RO)**ISP_AF_FDONE_INT_ST** The masked interrupt status of ISP_AF_FDONE_INT. (RO)**ISP_AF_ENV_INT_ST** The masked interrupt status of ISP_AF_ENV_INT. (RO)**ISP_AWB_FDONE_INT_ST** The masked interrupt status of ISP_AWB_FDONE_INT. (RO)**ISP_HIST_FDONE_INT_ST** The masked interrupt status of ISP_HIST_FDONE_INT. (RO)**ISP_FRAME_INT_ST** The masked interrupt status of ISP_FRAME_INT. (RO)**ISP_BLC_FRAME_INT_ST** The masked interrupt status of ISP_BLC_FRAME_INT. (RO)**ISP_LSC_FRAME_INT_ST** The masked interrupt status of ISP_LSC_FRAME_INT. (RO)

Continued on the next page...

Register 36.141. ISP_INT_ST_REG (0x0068)

Continued from the previous page...

ISP_DPC_FRAME_INT_ST The masked interrupt status of ISP_DPC_FRAME_INT. (RO)

ISP_BF_FRAME_INT_ST The masked interrupt status of ISP_BF_FRAME_INT. (RO)

ISP_DEMOSAIC_FRAME_INT_ST The masked interrupt status of ISP_DEMOSAIC_FRAME_INT. (RO)

ISP_CCM_FRAME_INT_ST The masked interrupt status of ISP_CCM_FRAME_INT. (RO)

ISP_GAMMA_FRAME_INT_ST The masked interrupt status of ISP_GAMMA_FRAME_INT. (RO)

ISP_RGB2YUV_FRAME_INT_ST The masked interrupt status of ISP_RGB2YUV_FRAME_INT. (RO)

ISP_SHARP_FRAME_INT_ST The masked interrupt status of ISP_SHARP_FRAME_INT. (RO)

ISP_COLOR_FRAME_INT_ST The masked interrupt status of ISP_COLOR_FRAME_INT. (RO)

ISP_YUV2RGB_FRAME_INT_ST The masked interrupt status of ISP_YUV2RGB_FRAME_INT. (RO)

ISP_TAIL_IDI_FRAME_INT_ST The masked interrupt status of ISP_TAIL_IDI_FRAME_INT. (RO)

ISP_HEADER_IDI_FRAME_INT_ST The masked interrupt status of ISP_HEADER_IDI_FRAME_INT.
(RO)

ISP_CROP_FRAME_INT_ST The masked interrupt status of ISP_CROP_FRAME. (RO)

ISP_WBG_FRAME_INT_ST The masked interrupt status of ISP_WBG_FRAME_INT. (RO)

ISP_CROP_ERR_INT_ST The masked interrupt status of ISP_CROP_ERR_INT. (RO)

Register 36.142. ISP_INT_ENA_REG (0x006C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	1	1

ISP_CROP_ERR_INT_ENA

ISP_WBG_FRAME_INT_ENA

ISP_CROP_FRAME_INT_ENA

ISP_HEADER_IDI_FRAME_INT_ENA

ISP_TAIL_IDI_FRAME_INT_ENA

ISP_YUV2RGB_FRAME_INT_ENA

ISP_COLOR_FRAME_INT_ENA

ISP_SHARP_FRAME_INT_ENA

ISP_RGB2YUV_FRAME_INT_ENA

ISP_GAMMA_FRAME_INT_ENA

(reserved)

ISP_CCM_FRAME_INT_ENA

ISP_DEMOSAIC_FRAME_INT_ENA

ISP_BF_FRAME_INT_ENA

ISP_DPC_FRAME_INT_ENA

ISP_LSC_FRAME_INT_ENA

ISP_BLC_FRAME_INT_ENA

ISP_FRAME_INT_ENA

ISP_HIST_FDONE_INT_ENA

ISP_AWB_FDONE_INT_ENA

ISP_AF_ENV_INT_ENA

ISP_AF_FDONE_INT_ENA

ISP_AE_FRAME_INT_ENA

ISP_AE_MONITOR_DONE_INT_ENA

ISP_GAMMA_XCOORD_DONE_INT_ENA

ISP_DPC_CHECK_DONE_INT_ENA

ISP_MIPI_HNUM_UNMATCH_INT_ENA

ISP_DATA_TYPE_SETTING_INT_ENA

ISP_HVNUM_SETTING_ERR_INT_ENA

ISP_BUF_FULL_INT_ENA

ISP_ASYNC_FIFO_OVF_INT_ENA

ISP_DATA_TYPE_ERR_INT_ENA

Reset

ISP_DATA_TYPE_ERR_INT_ENA Write 1 to enable ISP_DATA_TYPE_ERR_INT. (R/W)

ISP_ASYNC_FIFO_OVF_INT_ENA Write 1 to enable ISP_ASYNC_FIFO_OVF_INT. (R/W)

ISP_BUF_FULL_INT_ENA Write 1 to enable ISP_BUF_FULL_INT. (R/W)

ISP_HVNUM_SETTING_ERR_INT_ENA Write 1 to enable ISP_HVNUM_SETTING_ERR_INT. (R/W)

ISP_DATA_TYPE_SETTING_ERR_INT_ENA Write 1 to enable ISP_DATA_TYPE_SETTING_ERR_INT.
(R/W)

ISP_MIPI_HNUM_UNMATCH_INT_ENA Write 1 to enable ISP_MIPI_HNUM_UNMATCH_INT. (R/W)

ISP_DPC_CHECK_DONE_INT_ENA Write 1 to enable ISP_DPC_CHECK_DONE_INT. (R/W)

ISP_GAMMA_XCOORD_ERR_INT_ENA Write 1 to enable ISP_GAMMA_XCOORD_ERR_INT. (R/W)

ISP_AE_MONITOR_INT_ENA Write 1 to enable ISP_AE_MONITOR_INT. (R/W)

ISP_AE_FRAME_DONE_INT_ENA Write 1 to enable ISP_AE_FRAME_DONE_INT. (R/W)

ISP_AF_FDONE_INT_ENA Write 1 to enable ISP_AF_FDONE_INT. (R/W)

ISP AF ENV INT ENA Write 1 to enable ISP AF ENV INT. (R/W)

ISP_AWB_FDONE_INT_ENA Write 1 to enable ISP_AWB_FDONE_INT. (R/W)

ISP_HIST_FDONE_INT_ENA Write 1 to enable ISP_HIST_FDONE_INT. (R/W)

ISP_FRAME_INT_ENA Write 1 to enable ISP_FRAME_INT. (R/W)

ISP_BLC_FRAME_INT_ENA Write 1 to enable ISP_BLC_FRAME_INT. (R/W)

ISP_LSC_FRAME_INT_ENA Write 1 to enable ISP_LSC_FRAME_INT. (R/W)

ISP_DPC_FRAME_INT_ENA Write 1 to enable ISP_DPC_FRAME_INT. (R/W)

ISP_BF_FRAME_INT_ENA Write 1 to enable ISP_BF_FRAME_INT. (R/W)

ISP DEMOSAIC FRAME INT ENA Write 1 to enable ISP DEMOSAIC FRAME INT. (R/W)

Continued on the next page...

Register 36.142. ISP_INT_ENA_REG (0x006C)

Continued from the previous page...

ISP_CCM_FRAME_INT_ENA Write 1 to enable ISP_CCM_FRAME_INT. (R/W)

ISP_GAMMA_FRAME_INT_ENA Write 1 to enable ISP_GAMMA_FRAME_INT. (R/W)

ISP_RGB2YUV_FRAME_INT_ENA Write 1 to enable ISP_RGB2YUV_FRAME_INT. (R/W)

ISP_SHARP_FRAME_INT_ENA Write 1 to enable ISP_SHARP_FRAME_INT. (R/W)

ISP_COLOR_FRAME_INT_ENA Write 1 to enable ISP_COLOR_FRAME_INT. (R/W)

ISP_YUV2RGB_FRAME_INT_ENA Write 1 to enable ISP_YUV2RGB_FRAME_INT. (R/W)

ISP_TAIL_IDI_FRAME_INT_ENA Write 1 to enable ISP_TAIL_IDI_FRAME_INT. (R/W)

ISP_HEADER_IDI_FRAME_INT_ENA Write 1 to enable ISP_HEADER_IDI_FRAME_INT. (R/W)

ISP_CROP_FRAME_INT_ENA Write 1 to enable ISP_CROP_FRAME_INT. (R/W)

ISP_WBG_FRAME_INT_ENA Write 1 to enable ISP_WBG_FRAME_INT. (R/W)

ISP_CROP_ERR_INT_ENA Write 1 to enable ISP_CROP_ERR_INT. (R/W)

Register 36.143. ISP_INT_CLR_REG (0x0070)

ISP_CROP_ERR_INT_CLR																																ISP_WBG_FRAME_INT_CLR																																ISP_CROP_FRAME_INT_CLR																																ISP_HEADER_IDI_FRAME_INT_CLR																																ISP_TAIL_IDI_FRAME_INT_CLR																																ISP_YUV2RGB_FRAME_INT_CLR																																ISP_COLOR_FRAME_INT_CLR																																ISP_SHARP_FRAME_INT_CLR																																ISP_RGB2YUV_FRAME_INT_CLR																																ISP_GAMMA_FRAME_INT_CLR																																ISP_CCM_FRAME_INT_CLR																																(reserved)																																ISP_DEMOSAIC_FRAME_INT_CLR																																ISP_BF_FRAME_INT_CLR																																ISP_DPC_FRAME_INT_CLR																																ISP_LSC_FRAME_INT_CLR																																ISP_BLC_FRAME_INT_CLR																																ISP_FRAME_INT_CLR																																ISP_HIST_FDONE_INT_CLR																																ISP_AWB_FDONE_INT_CLR																																ISP_AF_ENV_INT_CLR																																ISP_AF_FDONE_INT_CLR																																ISP_AE_FRAME_DONE_INT_CLR																																ISP_GAMMA_XCOORD_ERR_INT_CLR																																ISP_DPC_CHECK_DONE_INT_CLR																																ISP_MIPI_HNUM_UNMATCH_INT_CLR																																ISP_DATA_TYPE_SETTING_ERR_INT_CLR																																ISP_HVNUM_SETTING_ERR_INT_CLR																																ISP_BUF_FULL_INT_CLR																																ISP_ASYNC_FIFO_OVF_INT_CLR																																ISP_DATA_TYPE_ERR_INT_CLR																																																																																										
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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Reset

ISP_DATA_TYPE_ERR_INT_CLR Write 1 to clear ISP_DATA_TYPE_ERR_INT. (WT)**ISP_ASYNC_FIFO_OVF_INT_CLR** Write 1 to clear ISP_ASYNC_FIFO_OVF_INT. (WT)**ISP_BUF_FULL_INT_CLR** Write 1 to clear ISP_BUF_FULL_INT. (WT)**ISP_HVNUM_SETTING_ERR_INT_CLR** Write 1 to clear ISP_HVNUM_SETTING_ERR_INT. (WT)**ISP_DATA_TYPE_SETTING_ERR_INT_CLR** Write 1 to clear ISP_DATA_TYPE_SETTING_ERR_INT. (WT)**ISP_MIPI_HNUM_UNMATCH_INT_CLR** Write 1 to clear ISP_MIPI_HNUM_UNMATCH_INT. (WT)**ISP_DPC_CHECK_DONE_INT_CLR** Write 1 to clear ISP_DPC_CHECK_DONE_INT. (WT)**ISP_GAMMA_XCOORD_ERR_INT_CLR** Write 1 to clear ISP_GAMMA_XCOORD_ERR_INT. (WT)**ISP_AE_MONITOR_INT_CLR** Write 1 to clear ISP_AE_MONITOR_INT. (WT)**ISP_AE_FRAME_DONE_INT_CLR** Write 1 to clear ISP_AE_FRAME_DONE_INT. (WT)**ISP_AF_FDONE_INT_CLR** Write 1 to clear ISP_AF_FDONE_INT. (WT)**ISP_AF_ENV_INT_CLR** Write 1 to clear ISP_AF_ENV_INT. (WT)**ISP_AWB_FDONE_INT_CLR** Write 1 to clear ISP_AWB_FDONE_INT. (WT)**ISP_HIST_FDONE_INT_CLR** Write 1 to clear ISP_HIST_FDONE_INT. (WT)**ISP_FRAME_INT_CLR** Write 1 to clear ISP_FRAME_INT. (WT)**ISP_BLC_FRAME_INT_CLR** Write 1 to clear ISP_BLC_FRAME_INT. (WT)**ISP_LSC_FRAME_INT_CLR** Write 1 to clear ISP_LSC_FRAME_INT. (WT)**ISP_DPC_FRAME_INT_CLR** Write 1 to clear ISP_DPC_FRAME_INT. (WT)**ISP_BF_FRAME_INT_CLR** Write 1 to clear ISP_BF_FRAME_INT. (WT)**ISP_DEMOSAIC_FRAME_INT_CLR** Write 1 to clear ISP_DEMOSAIC_FRAME_INT. (WT)**ISP_CCM_FRAME_INT_CLR** Write 1 to clear ISP_CCM_FRAME_INT. (WT)**ISP_GAMMA_FRAME_INT_CLR** Write 1 to clear ISP_GAMMA_FRAME_INT. (WT)

Continued on the next page...

Register 36.143. ISP_INT_CLR_REG (0x0070)

Continued from the previous page...

ISP_RGB2YUV_FRAME_INT_CLR Write 1 to clear ISP_RGB2YUV_FRAME_INT. (WT)

ISP_SHARP_FRAME_INT_CLR Write 1 to clear ISP_SHARP_FRAME_INT. (WT)

ISP_COLOR_FRAME_INT_CLR Write 1 to clear ISP_COLOR_FRAME_INT. (WT)

ISP_YUV2RGB_FRAME_INT_CLR Write 1 to clear ISP_YUV2RGB_FRAME_INT. (WT)

ISP_TAIL_IDI_FRAME_INT_CLR Write 1 to clear ISP_TAIL_IDI_FRAME_INT. (WT)

ISP_HEADER_IDI_FRAME_INT_CLR Write 1 to clear ISP_HEADER_IDI_FRAME_INT. (WT)

ISP_CROP_FRAME_INT_CLR Write 1 to clear ISP_CROP_FRAME_INT. (WT)

ISP_WBG_FRAME_INT_CLR Write 1 to clear ISP_WBG_FRAME_INT. (WT)

ISP_CROP_ERR_INT_CLR Write 1 to clear ISP_CROP_ERR_INT. (WT)

36.9.2 MIPI CSI_Bridge Registers

The addresses in this section are relative to **MIPI CSI_Bridge** base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 36.144. CSI_BRIG_CLK_EN_REG (0x0000)

Diagram of the 32-bit CS1_BRIG_CLK_EN register structure:

- Bit 31: (reserved)
- Bit 1: (reserved)
- Bit 0: (reserved)

Reset

CSI_BRIG_CLK_EN Configures whether the CSI_Bridge clock is always on.

0: The clock is on only during register access

1: The clock is always on

(R/W)

Register 36.145. CSI_BRIG_CSI_EN_REG (0x0004)

(reserved)																																CSI_BRIG_CSI_BRIG_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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CSI_BRIG_CSI_BRIG_EN Configures whether to enable CSI_Bridge.

0: Disable

1: Enable

(R/W)

Register 36.146. CSI_BRIG_BUF_FLOW_CTL_REG (0x000C)

(reserved)		CSI_BRIG_CSI_BUF_DEPTH														(reserved)		CSI_BRIG_CSI_BUF_AFULL_THRD															
31	30	29														16	15	14	13														0
0	0	0														0	0	2040														Reset	

CSI_BRIG_CSI_BUF_AFULL_THRD Configures the size threshold at which the CSI_Bridge buffer is about to be full. (R/W)

CSI_BRIG_CSI_BUF_DEPTH Represents the size of the CSI_Bridge buffer in use. (RO)

Register 36.147. CSI_BRIG_DMA_REQ_CFG_REG (0x0008)

(reserved)																CSI_BRIG_CSI_DMA_FLOW_CONTROLLER				(reserved)				CSI_BRIG_DMA_CFG_UPD_BY_BLK				CSI_BRIG_DMA_BURST_LEN				Reset
31															18	17	16					13	12	11					0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	128											

CSI_BRIG_DMA_BURST_LEN Configures the amount of 64-bit data in a VDMA burst transfer. (R/W)

CSI_BRIG_DMA_CFG_UPD_BY_BLK Configures when **CSI_BRIG_DMA_BURST_LEN** and **CSI_BRIG_DMABLK_SIZE** are updated.

0: Update upon completion of each VDMA block transfer

1: Update at the end of each frame transfer

(R/W)

CSI_BRIG_CSI_DMA_FLOW_CONTROLLER Configures the DMA flow controller.

0: DMA acts as the flow controller

1: CSI_bridge acts as the flow controller

(R/W)

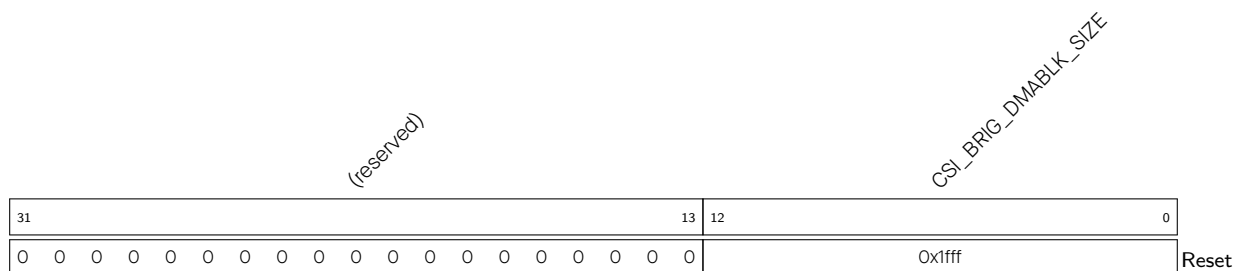
Register 36.148. CSI_BRIG_DMA_REQ_INTERVAL_REG (0x002C)

Diagram illustrating the structure of the **CSI_BRIG_DMA_REQ_INTERVAL** register. The register is 32 bits wide, divided into two 16-bit sections:

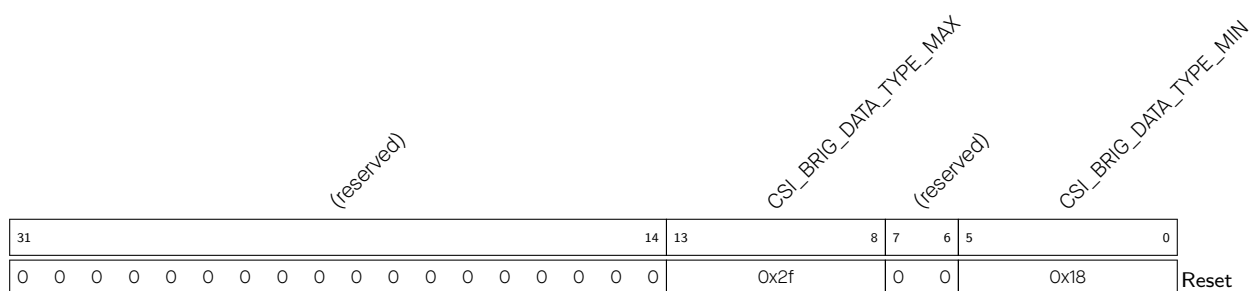
- Bits 31-16:** (reserved)
- Bits 15-0:** CSI_BRIG_DMA_REQ_INTERVAL

The value **0x01** is shown in the lower 16-bit section, and the **Reset** state is indicated on the right.

CSI_BRIG_DMA_REQ_INTERVAL Configures the interval of VDMA requests. 16'b1: 1 clock cycle, 16'b11: 2 clock cycles, 16'hFFFF: 16 clock cycles, etc. (R/W)

Register 36.149. CSI_BRIG_DMABLK_SIZE_REG (0x0030)

CSI_BRIG_DMABLK_SIZE Configures the number of VDMA bursts in a VDMA block transfer. (R/W)

Register 36.150. CSI_BRIG_DATA_TYPE_CFG_REG (0x0010)

CSI_BRIG_DATA_TYPE_MIN Configures the minimum value of DATA_TYPE for valid pixels. (R/W)

CSI_BRIG_DATA_TYPE_MAX Configures the maximum value of DATA_TYPE for valid pixels. (R/W)

Register 36.151. CSI_BRIG_FRAME_CFG_REG (0x0014)

(reserved)						CSI_BRIG_VADR_NUM_CHECK										CSI_BRIG_HAS_HSYNC_E										CSI_BRIG_HADR_NUM										CSI_BRIG_VADR_NUM									
31						26	25	24	23						12	11						0																							
0	0	0	0	0	0	0	1	480					480					Reset																											

CSI_BRIG_VADR_NUM Configures the height of an image frame. Measurement unit: line. (R/W)

CSI_BRIG_HADR_NUM Configures the amount of 64-bit data in the width of an image frame. (R/W)

CSI_BRIG_HAS_HSYNC_E Configures whether the Image Interface 64 input includes HSYNC_START and HSYNC_END packets.

0: Not included

1: Included

(R/W)

CSI_BRIG_VADR_NUM_CHECK Configures whether to enable line number checking.

0: Disable

1: Enable

(R/W)

Register 36.152. CSI_BRIG_ENDIAN_MODE_REG (0x0018)

[illegible]

CSI_BRIG_BYTE_ENDIAN_ORDER Configures the byte order of 64-bit data.

0: Byte order remains unchanged

1: Reverse the high and low bytes

(R/W)

Register 36.153. CSI_BRIG_INT_RAW_REG (0x001C)

(reserved)																								CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_RAW																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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CSI_BRIG_VADR_NUM_GT_INT_RAW The raw interrupt status of CSI_BRIG_VADR_NUM_GT_INT. (R/WTC/SS)

CSI_BRIG_VADR_NUM_LT_INT_RAW The raw interrupt status of CSI_BRIG_VADR_NUM_LT_INT. (R/WTC/SS)

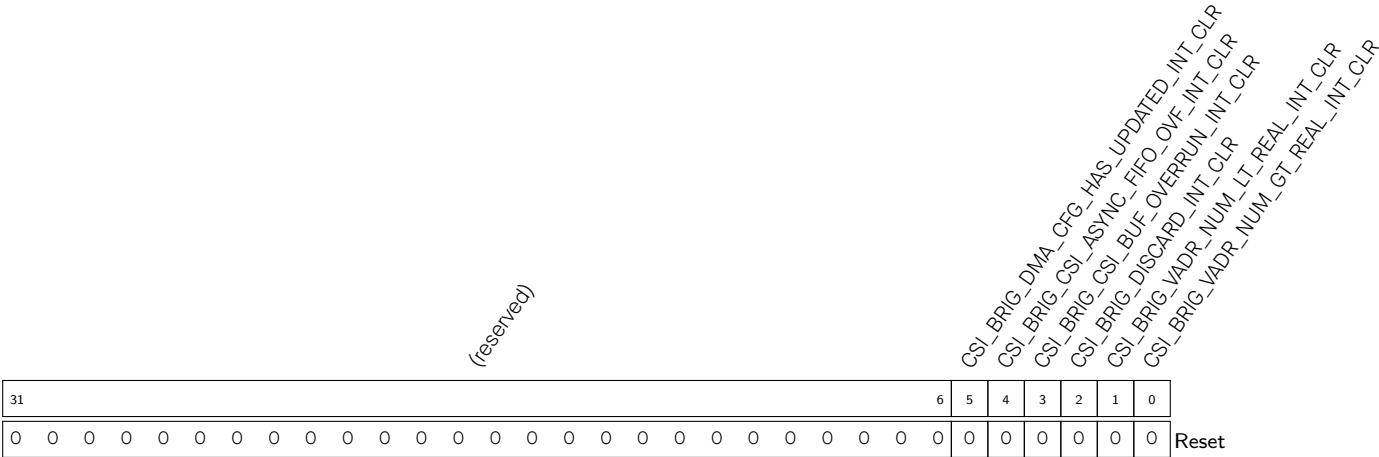
CSI_BRIG_DISCARD_INT_RAW The raw interrupt status of CSI_BRIG_DISCARD_INT. (R/WTC/SS)

CSI_BRIG_CSI_BUF_OVERRUN_INT_RAW The raw interrupt status of CSI_BRIG_CSI_BUF_OVERRUN_INT. (R/WTC/SS)

CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT_RAW The raw interrupt status of CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT. (R/WTC/SS)

CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_RAW The raw interrupt status of CSI_BRIG_DMA_CFG_HAS_UPDATED_INT. (R/WTC/SS)

Register 36.154. CSI_BRIG_INT_CLR_REG (0x0020)



- CSI_BRIG_VADR_NUM_GT_REAL_INT_CLR Write 1 to clear CSI_BRIG_VADR_NUM_GT_REAL_INT.
(WT)
- CSI_BRIG_VADR_NUM_LT_REAL_INT_CLR Write 1 to clear CSI_BRIG_VADR_NUM_LT_REAL_INT.
(WT)
- CSI_BRIG_DISCARD_INT_CLR Write 1 to clear CSI_BRIG_DISCARD_INT. (WT)
- CSI_BRIG_CSI_BUF_OVERRUN_INT_CLR Write 1 to clear CSI_BRIG_CSI_BUF_OVERRUN_INT. (WT)
- CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT_CLR Write 1 to clear CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT.
(WT)
- CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_CLR Write 1 to clear CSI_BRIG_DMA_CFG_HAS_UPDATED_INT.
(WT)

Register 36.155. CSI_BRIG_INT_ST_REG (0x0024)

Diagram of the 32-bit Reset register. The register is divided into two main sections. The upper 26 bits are labeled "(reserved)". The lower 6 bits are labeled "Reset" and are numbered 31 down to 0. Bit 31 is the MSB, and bit 0 is the LSB. The bits are: 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0. The bits 31 through 6 are reserved. The bits 5 through 0 are the Reset register bits.

CSI_BRIG_VADR_NUM_GT_INT_ST The masked interrupt status of CSI_BRIG_VADR_NUM_GT_INT.
(RO)

CSI_BRIG_VADR_NUM_LT_INT_ST	The masked interrupt status of CSI_BRIG_VADR_NUM_LT_INT.
(RO)	

CSI_BRIG_DISCARD_INT_ST The masked interrupt status of CSI_BRIG_DISCARD_INT. (RO)

CSI_BRIG_CSI_BUF_OVERRUN_INT_ST	The masked interrupt status of CSI_BRIG_CSI_BUF_OVERRUN_INT. (RO)
--	---

CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT_ST The masked interrupt status of CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT. (RO)

CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_ST	The masked interrupt status of CSI_BRIG_DMA_CFG_HAS_UPDATED_INT. (RO)
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Register 36.156. CSI_BRIG_INT_ENA_REG (0x0028)

(reserved)																								CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_ENA			
																								CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT_ENA			
																								CSI_BRIG_CSI_BUF_OVERRUN_INT_ENA			
																								CSI_BRIG_DISCARD_INT_ENA			
																								CSI_BRIG_VADR_NUM_LT_INT_ENA			
																								CSI_BRIG_VADR_NUM_GT_INT_ENA			
31																			6	5	4	3	2	1	0	Reset	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CSI_BRIG_VADR_NUM_GT_INT_ENA Write 1 to enable CSI_BRIG_VADR_NUM_GT_INT. (R/W)

CSI_BRIG_VADR_NUM_LT_INT_ENA Write 1 to enable CSI_BRIG_VADR_NUM_LT_INT. (R/W)

CSI_BRIG_DISCARD_INT_ENA Write 1 to enable CSI_BRIG_DISCARD_INT. (R/W)

CSI_BRIG_CSI_BUF_OVERRUN_INT_ENA Write 1 to enable CSI_BRIG_CSI_BUF_OVERRUN_INT.
(R/W)

CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT_ENA Write 1 to enable CSI_BRIG_CSI_ASYNC_FIFO_OVF_INT.
(R/W)

CSI_BRIG_DMA_CFG_HAS_UPDATED_INT_ENA Write 1 to enable
CSI_BRIG_DMA_CFG_HAS_UPDATED_INT. (R/W)

Register 36.157. CSI_BRIG_HOST_CTRL_REG (0x0040)

(reserved)																												CSI_BRIG_CSI_ENABLECLK	
31																												1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	Reset

CSI_BRIG_CSI_ENABLECLK Configures whether to enable the clock lane of CSI PHY.

0: Disable

1: Enable

(R/W)

Register 36.158. CSI_BRIG_HOST_CM_CTRL_REG (0x0048)

(reserved)																												CSI_BRIG_CSI_HOST_CM_8BIT_SWAP CSI_BRIG_CSI_HOST_CM_16BIT_SWAP CSI_BRIG_CSI_HOST_CM_LANE_NUM CSI_BRIG_CSI_HOST_CM_TX CSI_BRIG_CSI_HOST_CM_RX_YUV422_FORMAT CSI_BRIG_CSI_HOST_CM_RX_RGB_FORMAT CSI_BRIG_CSI_HOST_CM_RX CSI_BRIG_CSI_HOST_CM_BYPASS CSI_BRIG_CSI_HOST_CM_EN							
31																14		13	12	11	10	9	8	7	6	4		3	2	1	0	Reset			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0x0	0x0	0x0	0x0	1	1								

CSI_BRIG_CSI_HOST_CM_EN Configures whether to enable CSI CM.

0: Disable. In this case, no data is output from CSI CM, and the downstream ISP cannot receive data.

1: Enable

(R/W)

CSI_BRIG_CSI_HOST_CM_BYPASS Configures whether to bypass CSI CM.

0: Do not bypass

1: Bypass

(R/W)

CSI_BRIG_CSI_HOST_CM_RX Configures the input image format of CSI CM.

0: RGB888

1: RGB565

2: YUV422

3: YUV420

(R/W)

CSI_BRIG_CSI_HOST_CM_RX_RGB_FORMAT Configures the channel order of input RGB data.

0: RGB

1: BGR

2: RGB

3: BRG

4: GRB

5: GBR

Others: Invalid

(R/W)

Continued on the next page...

Register 36.158. CSI_BRIG_HOST_CM_CTRL_REG (0x0048)

Continued from the previous page...

CSI_BRIG_CSI_HOST_CM_RX_YUV422_FORMAT Configures the channel order of input YUV422 data.

0: YVYU

1: YUYV

2: VYUY

3: UYVY

(R/W)

CSI_BRIG_CSI_HOST_CM_TX Configures the output image format of CSI CM.

0: RGB888

1: RGB565

2: YUV422

3: YUV420

(R/W)

CSI_BRIG_CSI_HOST_CM_LANE_NUM Configures the number of CSI channels; valid only when both input and output formats are RGB888.

0: 1 channel

1: 2 channels

(R/W)

CSI_BRIG_CSI_HOST_CM_16BIT_SWAP Configures whether to enable 16-bit high/low swapping.

0: Disable

1: Enable

(R/W)

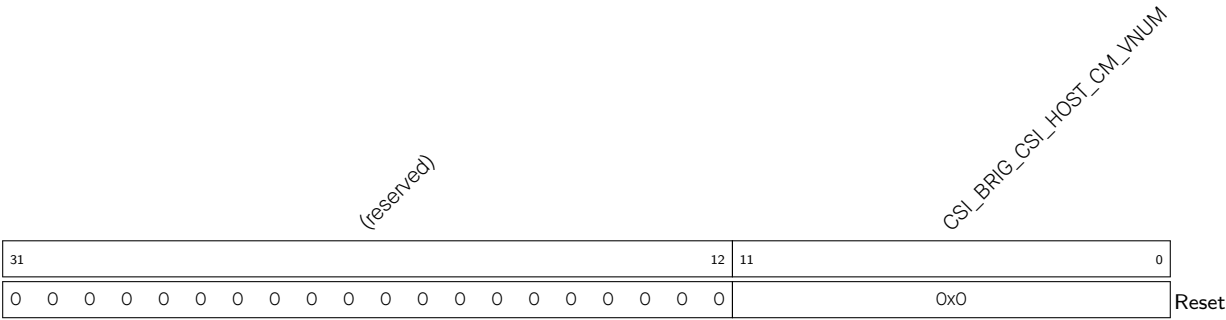
CSI_BRIG_CSI_HOST_CM_8BIT_SWAP Configures whether to enable 8-bit high/low swapping.

0: Disable

1: Enable

(R/W)

Register 36.159. CSI_BRIG_HOST_SIZE_CTRL_REG (0x004C)



CSI_BRIG_CSI_HOST_CM_HNUM Configures the horizontal size of the MIPI CSI input image interface (in units of 32-bit data). The value is $\text{line_pix_num} * \text{bits_per_pix} / 32 - 1$, and is valid only when converting from YUV422 to YUV420. (R/W)